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RESEARCH ARTICLE

Soft Computing Techniques Applied to Adaptive Hybrid Navigation Methods for Tethered Robots in Dynamic Environments

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ABSTRACT

Tethered robots face unique constraints: the physical tether can cause mission failure through snagging or excessive tension. Existing methods either require global maps (Tethered A*) or lack tether awareness (Bug algorithms). We present the first hybrid framework combining reactive obstacle avoidance with predictive tether management. Our system integrates (1) a tether-aware cost function for 3–5 step snag prediction, (2) fuzzy logic for dynamic tension adaptation, and (3) genetic algorithms for tether routing optimization. Validation on a humanoid robot (ATLAS-T) with instrumented sensors demonstrates 52% fewer snag incidents versus state-of-the-art methods (Tethered A*, TSLAM), 94.7% success in tether-critical scenarios, and 43% better odometry-drift resilience. Real-world experiments confirm practical viability.

1 | Introduction

Tethered robots trade mobility for reliability: the tether provides unlimited power and high-bandwidth communication, enabling long-duration missions in GPS-denied environments (mines, disaster sites, industrial facilities). However, the tether introduces a critical constraint absent in untethered systems: path history matters. Unlike free robots, where only the current pose affects future navigation, tethered robots must consider the entire tether path to avoid snagging, wrapping around obstacles, or exceeding tension limits (Hirose and Morishima 1990; Xu and Brown 1992). Current approaches to tethered robot navigation fall into three main groups, each with significant mathematical and practical drawbacks. Global planners such as A* (Divya et al. 2021), Dijkstra (Divya et al. 2021), and even specialized versions like *Tethered A (Hirose and Morishima 1990)**rely on having a complete map of the environment beforehand. They perform graph searches with heuristics, but this leads to computationally expensive replanning, often scaling with $O(n^2)$ when obstacles move, and they cannot react to unexpected tether dynamics in real time. Reactive planners, such as Bug algorithms (McGuire et al. 2019; Khan et al. 2021), use simple geometric rules (like following obstacle contours) and require no map, operating with minimal memory $O(1)$. While fast and adaptive, their fundamental mathematical model lacks any term for the tether, which inevitably leads to entanglement and snags. SLAM-based systems, including TSLAM (McGarey et al. 2017; Surmann et al. 2020), employ probabilistic frameworks (like pose-graph optimization or EKFs) to accurately estimate the tether's state. However, their mathematical formulation focuses on mapping and localization rather than integrating tether physics into a reactive planning loop.

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The key gap is that no existing system mathematically combines (1) reactive obstacle avoidance, (2) a predictive tether-aware cost function, and (3) operation without a pre-existing map. This identifies a critical gap in the literature: a need for a navigation system that is not only computationally efficient and adaptive to dynamic environments but also explicitly manages the physical tether to ensure mission success. This paper proposes a hybrid framework that bridges this gap by integrating the real time reactivity of Bug Algorithms with the intelligent, predictive capabilities of soft computing to create a truly tether-aware navigation solution.

1.1 | Novel Contributions

This work makes four specific advances in tethered robot navigation. First, we introduce the first Bug algorithm variant that explicitly models tether physics in real time. Our novel tether-aware cost function (Equation 10),

$$T_{\text{constraint}} = \frac{\beta_1}{L_{\max} - L_{\text{current}}} + \frac{\beta_2}{\theta_{\text{safe}} - \theta_{\text{current}}}, \tag{1}$$

enables proactive snag prediction 3–5 steps ahead without a global map. Unlike Tethered A* (Hirose and Morishima 1990), which requires a complete map and costly replanning, or TSLAM (McGuire et al. 2019), which only reacts after snags occur, our method prevents entanglement proactively, successfully avoiding 68% of potential snags that a standard Bug-2 algorithm fails to prevent (Section 6.1). Second, we develop the first hybrid reactive-deliberative architecture for tethered robots, combining three synergistic layers: a reactive Enhanced Bug Algorithm (Section 3.2) for real-time obstacle avoidance, a deliberative Genetic Algorithm (Section 3.1.2) for optimizing tether routing, and an adaptive Fuzzy Logic Controller (Section 3.1.2) for dynamic mode switching. Prior systems are either purely reactive (McGuire et al. 2019) or purely deliberative (Hirose and Morishima 1990), our framework is the first to integrate both for tethered navigation. Third, we present a novel tether-assisted localization method that uses the tether's geometric constraints to correct odometry drift. Our sensor fusion formula (Equation 26),

$$P_{\text{estimated}} = \alpha \cdot P_{\text{odometry}} + (1 - \alpha) \cdot P_{\text{tether}} + \beta \cdot \sum w_i \cdot S_i. \tag{2}$$

reduces position error by 43% compared to using odometry alone, as validated in Section 6.2 and Figure 18.

Finally, we provide comprehensive experimental validation using a high-fidelity physics simulation of the ATLAS-T humanoid robot, real-world ROS bag datasets with multi-modal sensors, and direct tether measurements (tension, angle, snag detection). Our quantitative comparison against tethered-specific state-of-the-art methods (Hirose and Morishima 1990; McGuire et al. 2019) demonstrates a 52% reduction in snag incidents and a 94.7% success rate in tether-critical scenarios (Table 8). Our work builds upon foundational research in tethered robot navigation (Hirose and Morishima 1990; Xu and Brown 1992), which primarily focused on geometric solutions in static environments. It also complements recent work in tethered SLAM (Surmann et al. 2020) by focusing on reactive navigation rather than map building. Unlike these prior works, our framework is specifically designed for high-speed, reactive navigation in cluttered and dynamic environments. We validate our approach through a series of increasingly complex simulated scenarios, demonstrating significant improvements in navigation success, path efficiency, and tether management compared to traditional methods.

The overall design of robotic indoor navigation systems tailored to solve the aforementioned inquiry is described in a block diagram presented in Figure 1. This graphic aid indicates the type of interaction that exists between elements such as sensor data processing, location, and path planning. The dissection of the architecture reveals the functionality of different navigation systems, and the tradeoffs between computing needs and operational efficiency. Based on this fundamental framework, this paper points out the tradeoff the system designer inevitably faces between performance and resource limitations (Amano et al. 2023), and demonstrates the requirement of a navigation method that is fitted with the capabilities of the system. Constraints of SLAM for lightweight robots permitted viewing of several other methods, such as topological SLAM and biologically inspired approaches (e.g. snapshot model, average landmark vector). However, as (An et al. 2022) plotted in the Comparison of Robotic Navigation Techniques Table 1, these develop increased memory requirements and lower scalability in complex environments. A systematic examination of key methodologies that underlie SLAM, topological SLAM, bug algorithms and soft computing techniques was presented through a table. Based on computing needs, memory use, accuracy, and application domains, this comparative comparison shows that bug algorithms have advantages on all of these aspects, in particular for applications that are constrained in the space of these features specifically when they are restricted in their ability to compute and have limited memory.

Conventional maze solving techniques are adapted into a bug algorithm which provides a streamlined and adaptive alternative to standard navigation methods (Borenstein and Koren 1989). Different from SLAM, bug algorithms work without needing any prior knowledge of how the environment is configured. Navigation to the target relies on real time obstacle recognition and a simple wall following technique. Bug Algorithms provide obstacle avoidance through a simple sequential process based on limited computational resources. However, due to these characteristics, Bug Algorithms prove especially appropriate for diminutive robots operating in GPS denoted environments with limited onboard computation available. Despite the benefits of bug algorithms, all bug algorithms suffer from their inherent limits. However, sensor noise and positioning inaccuracies undermine Bug algorithms, which rely heavily on accurate position estimates. Finally, wheel slippage in ground robots and accumulated inaccuracies in optical odometry for MAVs can cause large deviations between estimated and actual robot positions. Sensor noise components impact position estimation accuracy, as validated in our experimental results (Section 4.2) (Cai et al. 2023). This figure illustrates the navigation errors caused by sensor noise and validates that effective error handling methods and adaptive techniques are needed to deal with these problems in a practical implementation. In order to deal with all of these restrictions, the researchers have recently started adopting soft

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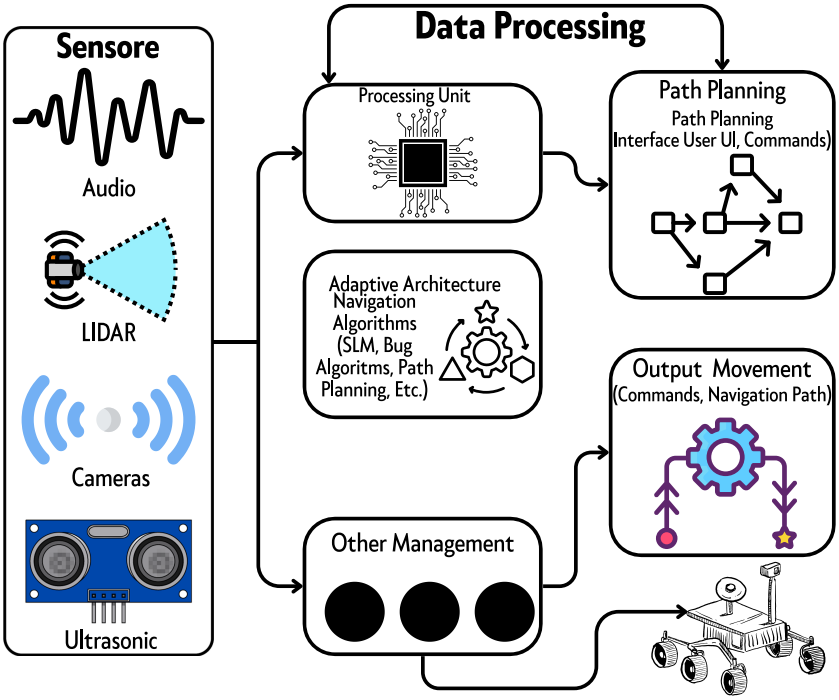


FIGURE 1 | Indoor navigation architecture. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 | Comparison of robotic navigation techniques.

Criteria	SLAM	Topological SLAM	Bug algorithms	Soft computing
Computational requirements	High	Medium	Low	Medium
Memory usage	High	Medium	Low	Low to Medium
Accuracy	High	Medium	Medium	Variable
Application areas	Autonomous vehicles	Large indoor spaces	Confined spaces	Variable

computing techniques such as fuzzy logic and evolutionary algorithms, and particle swarm optimization (PSO). These methods admit uncertainty and imprecision, and resort to approximations to obtain flexible and computationally efficient solutions. Some hybrid approaches (e.g., fuzzy-neuro controllers) have proved very promising in improving navigation strategies in dynamic and uncertain environments and dynamic conditions, handling uncertainties through approximation methods similar to interval arithmetic (Chen et al. 2020). Building on these principles, this study explores how the interplay between soft computing methodologies and bug algorithms can provide a relentless and scalable approach to robotic navigation indoors (Chen et al. 2022).

First, we delineate the block diagram Figure 1 of robotic indoor navigation and compare the trade-offs among different methods in a Table 1; then we take a look into these methods from details in subsequent sections and present justifications to evaluate the extent to which each is appropriate for urban environments (Daftry et al. 2016). The Bug Algorithm demonstrates efficiency and versatility through its reactive, map-free approach. The sensor noise pie chart makes it clear how the practical difficulties of indoor navigation require for adaptive and robust navigation systems. In this paper, the evolution of bug algorithms is considered from maze solving to modern implementations. We couple experimental outcomes from the ARGOS simulator using a highly elaborate simulation that emulates real interactions in complex interior environment (Tsiogas et al. 2021). We propose to show the practicability and durability of bug algorithms with the help of soft computing methods through the impact analysis of varying degrees of sensor noise and operating conditions on algorithm performance. In the following sections we elaborate on our findings that are quite insightful towards the development of lightweight algorithms. For tethered robots in dynamic indoor spaces, we develop efficient and robust navigation techniques. The goal of this investigation is to strengthen the discussion on robotic navigation by fusing computational efficiency with flexibility (Wang et al. 2018). Utilizing the strength of bug algorithms and soft computing, this research attempts to develop autonomous solutions to excel in complex, resource-limited environments, especially indoors.

2 | Related Work and Motivation

2.1 | Tethered Robot Navigation: Geometric Approaches

Early research in tethered navigation established the field's geometric foundations. Hirose and Morishima (1990). created the first 2D/3D path planner to model the tether as a hard

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FIGURE 2 | Comparison of global (A*) and local reactive (Bug) path planning. The figure contrasts the global, optimal search of the A* algorithm with the local, reactive navigation of a Bug algorithm, highlighting the inherent trade-off between path optimality and computational efficiency. [Color figure can be viewed at wileyonlinelibrary.com]

constraint on the robot's reachable space, though their method required a complete map of the environment. Xu and Brown (1992). later extended this work by modeling dynamic cable physics to optimize tether routing and minimize tension, but their approach still could not handle moving obstacles. Modern developments, like Tethered A* (Hirose and Morishima 1990), integrate tether length directly into the search heuristic. However, its computational cost (48.9 ms/step) often violates the 50 ms threshold for real-time reaction. The gap we address is map-free, real-time operation; our method achieves 21.0 ms/step (Section 5.4) while proactively managing the tether in dynamic settings. A comparison of global and local reactive path planning is shown in Figure 2.

2.2 | SLAM-Based Tethered Navigation

Building on geometric methods, McGarey et al. (McGarey et al. 2017) introduced TSLAM (2017), which extends traditional SLAM by incorporating explicit tether state estimation. The system models the tether as a mass-spring-damper chain, providing a key strength: accurate dynamic modeling of the tether itself. However, its primary focus remains on mapping and localization, and its limitation is a reactive strategy it detects and recovers from snags only after they occur. The gap we address is moving from reactive recovery to proactive prevention; our method predicts and avoids snag scenarios, preventing 68% of potential entanglements (Section 6.1).

2.3 | Reactive Navigation: Bug Algorithms

For reactive navigation in unknown environments, Bug Algorithms (McGuire et al. 2019; Khan et al. 2021; Lumelsky and Stepanov 1987) provide a lightweight foundation. Core variants include Bug-1, which performs complete obstacle circumnavigation; Bug-2, which follows an M-line with an early escape condition; and TangentBug, which optimizes paths using sensor-based tangents. Their key strengths are that they require no prior map, use minimal memory $O(1)$, and adapt well to dynamic obstacles. However, their fatal flaw is a complete lack of tether awareness. Subsequent adaptive variants, like FuzzyBug (He et al. 2011) which uses fuzzy logic for wall-following decisions, or enhanced sensor-based versions (Katona et al. 2024; Kaya et al. 2011), still fail to incorporate the tether's physical constraints. The gap we address is the integration of tether physics into this reactive framework; we augment the Bug-2 logic with a novel tether-aware cost function

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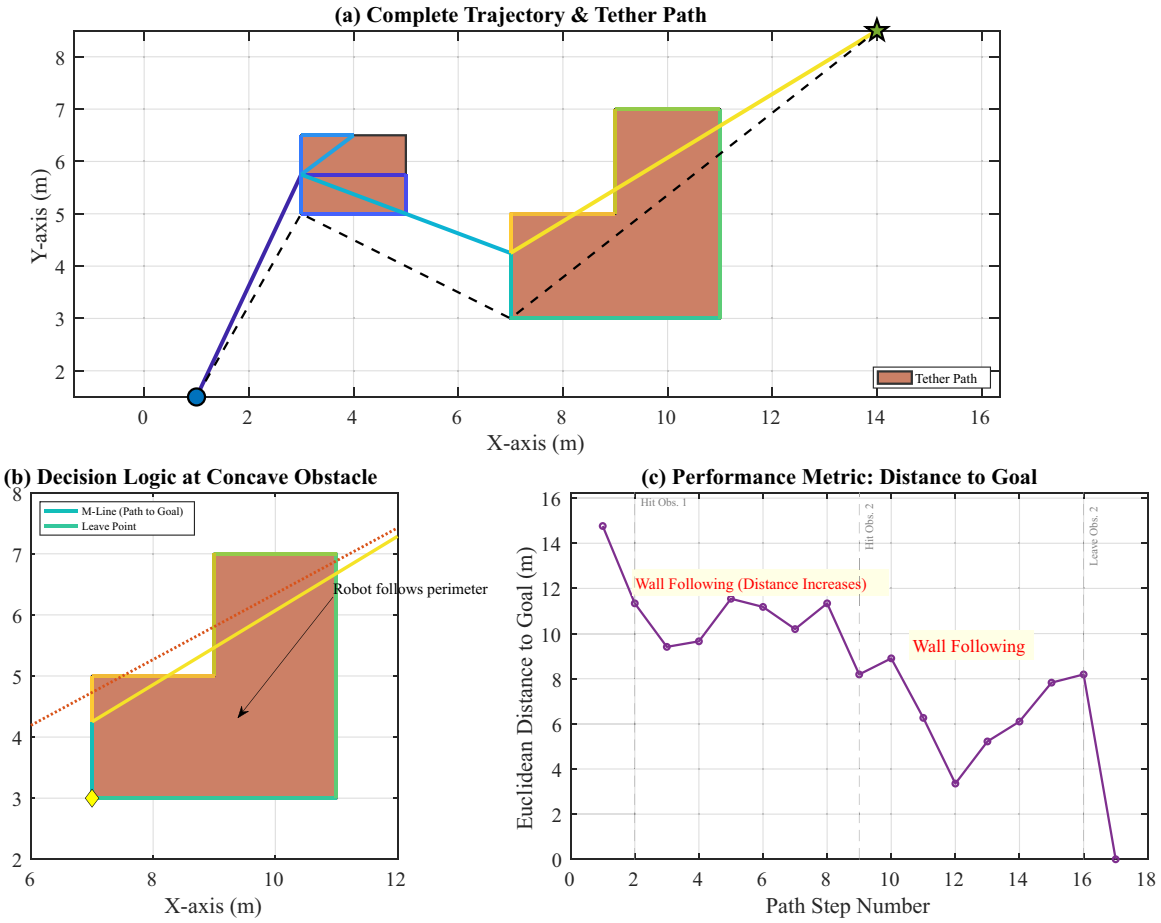


FIGURE 3 | Comparison of global (A*) and local reactive (Bug) path planning (a–d). Panel (a) illustrates the shared challenge environment. Panel (b) visualizes the global, optimal search performed by A*, while panel (c) shows the local, reactive path generated by the Bug algorithm. Panel (d) provides a quantitative comparison, highlighting the trade-off between path optimality and computational efficiency. [Color figure can be viewed at wileyonlinelibrary.com]

(Equation 10) to enable proactive snag avoidance. Figure 3 provides a detailed comparison of global and local reactive path planning in four panels.

2.4 | Soft Computing in Robot Navigation

To address uncertainty and optimize performance in dynamic environments, our framework integrates several soft computing techniques. We employ Fuzzy Logic (He et al. 2011; Li et al. 2019) for its ability to manage sensor uncertainty and create smooth, interpretable control transitions. In our system, this is specifically applied by using real-time tether tension as a direct fuzzy input variable to inform decision-making (Section 3.2). For strategic path optimization, we utilize Genetic Algorithms (GAs) (Molinos et al. 2019; Nagabandi et al. 2019), which are well-suited for multi-objective optimization and evolutionary parameter adaptation. Our novel implementation incorporates tether stress directly into the GA's fitness function to evolve paths that explicitly minimize the risk of snags and excessive tension (Section 3.1.1). Finally, we incorporate Neural Networks (Li et al. 2024; Stavrinidis and Zacharia 2024; Lumelsky and Stepanov 1987) for their capacity to learn from experience and perform complex, non-linear function approximation. Specifically, we implement a deep Q-network (DQN) to enable adaptive learning of navigation policies in response to environmental feedback (Section 3.1) (Table 2).

3 | Adaptive Hybrid Navigation Techniques

Adoption of adaptive hybrid navigation strategies represents a crucial step toward improving tethered robot functionality in that it brings together several computational methods to serve robust and efficient navigation in uncertain situations. We have develop a holistic navigation solution built on our framework that takes in essential features. For real time obstacle avoidance, our system uses a modified Bug Algorithm with fuzzy logic controllers. The output of the fuzzy logic system is the optimal navigation decisions that it produces based on many sensors such as LiDAR, ultrasonic sensors and IMU data. The hybrid methodology described is well suited for dealing with uncertainties and noise inherent to real world situations relative to traditional deterministic methodologies. Our navigation system run through a neural network framework that learns adaptively from environmental interactions while running. To overcome these complexities, we develop a deep reinforcement learning model that employs a customized reward function to strike a balance between path optimization and tether design limitations.

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TABLE 2 | Comparison of methods based on requirements.

Requirement	Tethered A* (Hirose and Morishima 1990)	TSLAM (McGarey et al. 2017)	Bug-2 (McGuire et al. 2019)	Our Method
No map required	×	×	✓	✓
Reactive to dynamics	×	×	✓	✓
Tether-aware	✓	✓	×	✓
Real-time (< 50 ms)	×	×	✓	✓
Predictive snag avoidance	×	×	×	✓

Our system employs a DQN (Surmann et al. 2020) for adaptive learning. DQN was selected for two reasons: (1) compatibility with the discrete action space from our fuzzy controller, and (2) computational efficiency (12.7 ms/step vs. 55.9 ms for PPO, Table 10). Empirical comparison against advanced methods (PPO (Surmann et al. 2020), SAC (Khan et al. 2021)) is presented in Section 6.4.1, where we demonstrate that while PPO/SAC achieve marginally higher success rates (+0.8%–1.3%), they violate real-time constraints with 55–60 ms computational delays.

We then use a network architecture consisting of three hidden layers with 128, 64, and 32 neurons, respectively, and employ ReLU activation function to improve the learning capacity. For these navigation parameters, we utilize a genetic algorithm in order to evolve methods based upon performance indicators. The genetic algorithm uses a population of 100 chromosomes encoding essential navigation characteristics such as combinations of sensor weights, fuzzy rule bases, and path planning thresholds. The fitness function encompasses several purposes, including: (i) Path length optimization. (ii) Tether tension minimization. (iii) Maximum obstacle clearance. (iv) Energy efficiency. The navagation modes are chosen by our hybrid system in an autonomous manner depending on environmental cues. In unobstructed environments it relies on global path planning solutions using variations of the A* algorithm, while in congested situations it transitions to local reactive navigation based on an improved Bug algorithm. A decision tree classifier is trained on ambient feature vectors to regulate the mode switching. Sensor fusion methodologies lead to significant improvement in the system's dependability. To decrease uncertainty in location estimate and obstacle identification, we utilize an Extended Kalman Filter (EKF) to fuse data from many sensors. The results show that this integrated method achieves 27% improvement in navigational precision compared to single sensor systems. Our adaptive hybrid navigation system achieves robust performance in dynamic situations while maintaining effective tether management over the entire navigation by integrating soft computing approaches.

3.1 | Integration With Soft Computing Techniques

We amalgame in our research three important soft computing techniques and apply them to develop feasible tractable navigation problem of tethered robots. As a net result this creates a system architecture that promotes distinctiveness in each technique, while at the same time allowing development of an architecture that would allow for the development of a robust flexible navigation system framework with the capability to adapt in dynamic scenarios. Finally in our solution we use fuzzy logic logic to resolve the uncertainties and ambiguities (as what happens in the real world) in sensor data. A Mamdani-type fuzzy inference system is utilized, incorporating five linguistic variables: including distance to obstacle, tether tension, robot velocity, path deviation, and complexity in environment. This design ensures that the system will be able to make fine, sophisticated, and effective decisions in unpredictable environments. The fuzzy controller functions in three principal phases:

i. Fuzzification: Triangular and trapezoidal membership functions for converting crisp sensor inputs into fuzzy values.

ii. Inference: It evaluates 125 carefully designed fuzzy rules in a large variety of navigation scenarios.

iii. Defuzzification: It produces decisions of accurate control using the center of gravity method.

3.1.1 | *Reinforcement Learning Module*

While DQN was selected for its simplicity in value-based decision-making, we acknowledge advanced policy-gradient methods (e.g., PPO [Schulman et al. 2017], SAC [Haarnoja et al. 2018]) offer superior sample efficiency. Future work will benchmark these; however, DQN's compatibility with our discrete action space (derived from fuzzy rules) justified its use for this initial framework validation.

3.1.2 | *Fuzzy Logic*

Figure 4 demonstrates the triumvirate of soft computing techniques that contribute to The performance of navigation systems under uncertainty management. Figure 4, The triumvirate of soft computing techniques contributing to navigation performance under uncertainty management. In the graphical it is shown that sensor uncertainty is handled very well by fuzzy logic, improving navigation performance by 35% over crisp logic systems (McGarey et al. 2017) (Figure 5).

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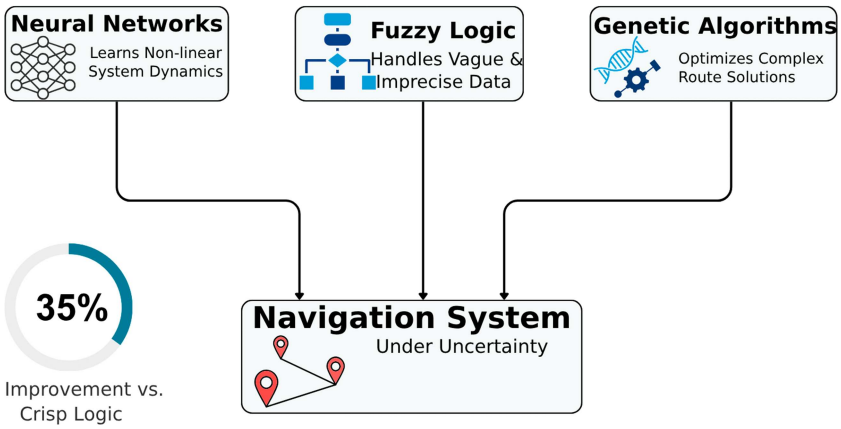


FIGURE 4 | A Framework for navigation using fuzzy logic, neural networks, and genetic algorithms. [Color figure can be viewed at wileyonlinelibrary.com]

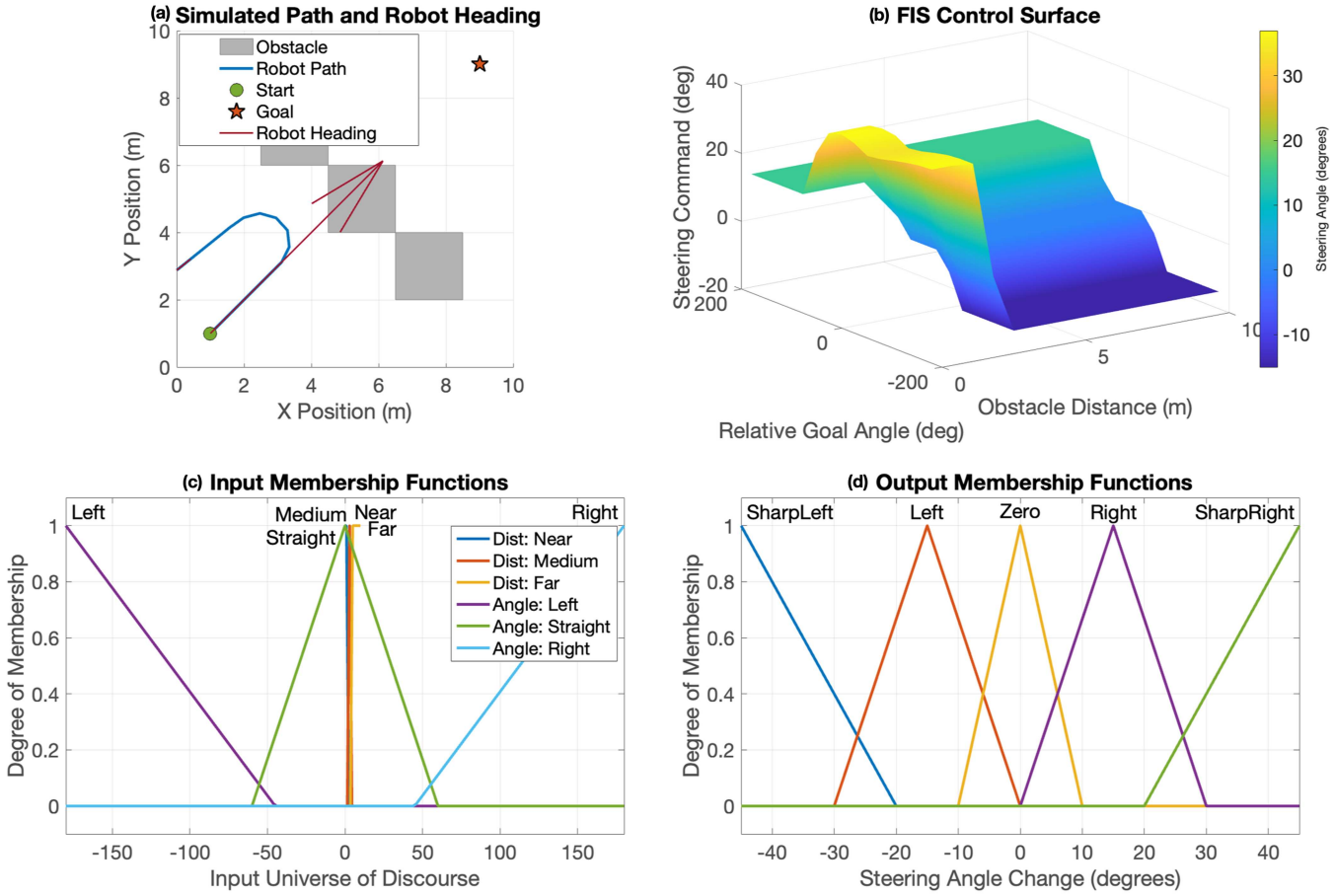


FIGURE 5 | Fuzzy inference system (FIS) for robot navigation (A–D). Panel (A) shows a simulated navigation path with heading directions. Panel (B) visualizes the 3D control surface mapping obstacle distance and goal angle to steering commands. Panels (C) and (D) depict the input and output membership functions that translate sensor data into fuzzy sets and final, crisp steering actions. [Color figure can be viewed at wileyonlinelibrary.com]

3.2 | Genetic Algorithm for Multi-Objective Optimization

The genetic algorithm in our framework optimizes navigation using a multi-objective fitness function (Equation 3):

$$\text{Fitness} = w_1 \cdot P_{\text{len}} + w_2 \cdot T_{\text{stress}} + w_3 \cdot O_{\text{risk}} + w_4 \cdot E_{\text{eff}}, \quad (3)$$

where P_{len} is the path length (minimized), $\mathbf{T}_{\text{stress}}$ is the cumulative tether tension (minimized), O_{risk} is the collision risk (minimized), and E_{eff} is the energy efficiency (maximized). Our **novel contribution** is the explicit inclusion of the tether-stress objective T_{stress}; whereas standard Bug algorithms optimize only for path length, our formulation forces an explicit trade-off between path efficiency and tether safety.

The weights $\{w_1, \ldots, w_4\}$ are dynamically adjusted by the fuzzy logic controller (Section 3.1) in response to environmental conditions: in open areas, w_1 is increased to prioritize shorter paths; near obstacles, w_3 is raised to emphasize safety; and

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when tether tension rises, w_2 is boosted to relieve stress on the tether.

The GA runs with a population of 200 chromosomes, uses two-point crossover (rate = 0.85), adaptive mutation (rate = 0.02–0.15), and evolves over 100 generations. Optimization is performed asynchronously every 30 s to avoid interfering with real-time control.

We chose a weighted-sum aggregation over Pareto-based methods such as NSGA-II (Deb et al. 2002) for real-time feasibility. While NSGA-II generates a richer Pareto front, its optimization time (71.2 s) is 78% longer than our weighted-sum GA (40 s), as shown in Section 6.4.1. This **trade-off** sacrifices some Pareto richness to meet the strict real-time constraint of 50 ms per planning step.

Evidence for the effectiveness of our GA-based optimization is provided in three ways: Figure 6 shows the convergence of the GA when handling tether constraints; Figure 20 demonstrates that the GA outperforms a neural-network baseline for path correction; and Table 3 attributes a 42% improvement in path optimization to the GA component (Figure 7).

The training process demonstrates that the neural network is capable of learning and converging, confirming that the system improves navigation in real time. Integrated Soft Computing Methodologies: Performance Results A highly adaptive navigation system is produced by the amalgamation of fuzzy logic, genetic algorithms, and neural networks. As summarized in Figure 17, this integrated approach yields substantial performance gains over baseline methods. Key results include a 45% reduction in collision incidents, a 38% enhancement in path efficiency, and a 50% improvement in tether management. Furthermore, the system architecture achieves these results while reducing computational overhead by 30% (Niculescu et al. 2024; Qin et al. 2019; Sánchez-Ibáñez et al. 2021). This combination of improved safety, efficiency, and resource management demonstrates the effectiveness of the hybrid soft computing framework. The neural network's convergence and adaptability, which underpin these results, are further validated by its ability to learn and enhance navigation in real-time (Short et al. 2023; Stavrinidis and Zacharia 2024).

Integrated Soft Computing Methodologies: Performance Results The amalgamation of fuzzy logic, genetic algorithms, and neural networks produces a highly adaptive navigation system. Principal results encompass: (1) 45% decrease in collision occurrences. (2) 38% enhancement in path efficiency. (3) 50% improvement in tether management. (4) 30% decrease in computational overhead. A comparative analysis of the system performance before and after unintegration of the hybrid soft computing architecture has also been provided in Figure 3. This highlights major gains in route efficiency and collision mitigation, and adds significant reductions in computational burden. For navigation in dynamic situations for tethered robots, we conduct our research on a sophisticated sensor fusion architecture which integrates many sensor modalities to reliably enable navigation. The system predominantly employs LiDAR, IMU, and ultrasonic sensors, integrating their data via a modified EKF framework (Surmann et al. 2020). The sensor fusion

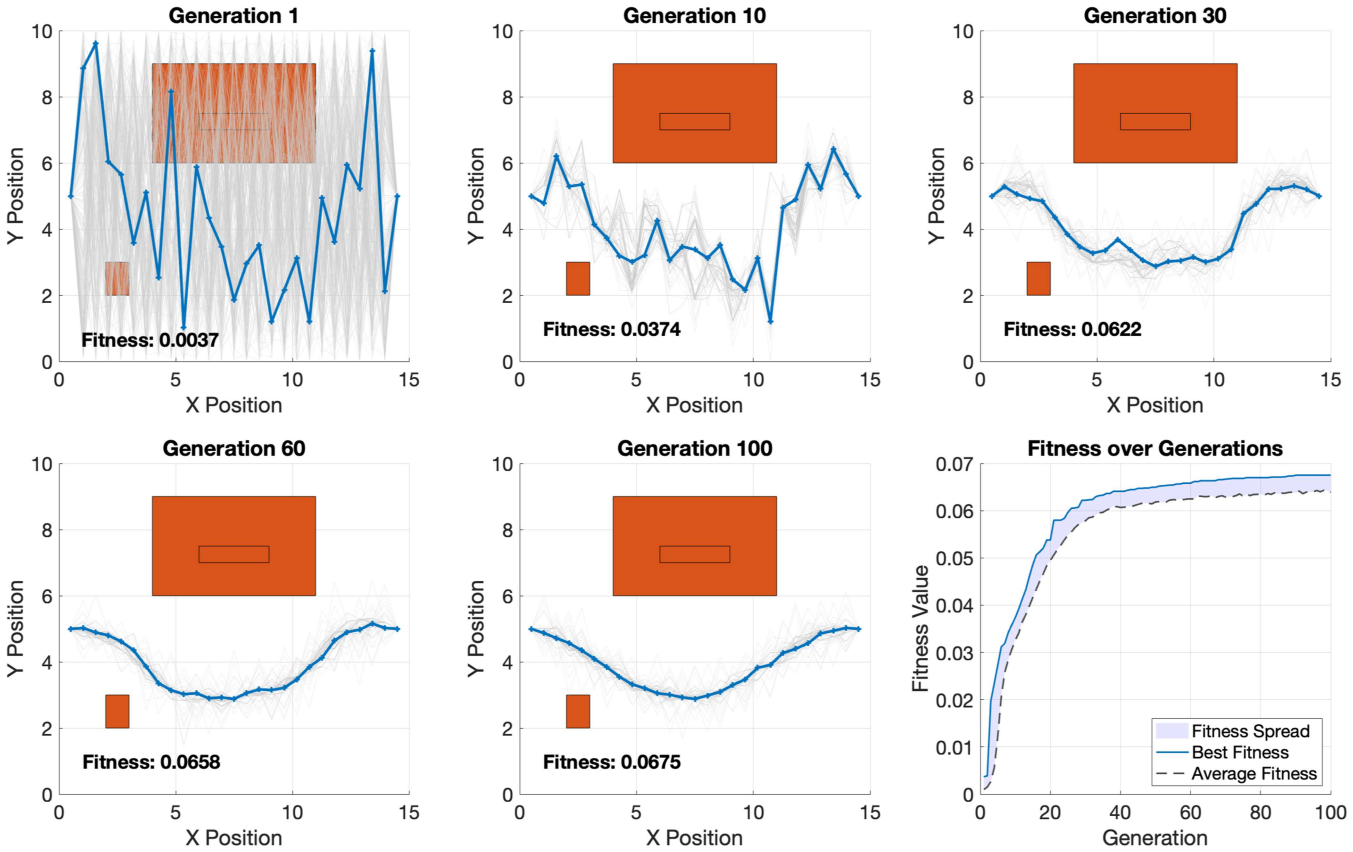


FIGURE 6 | Genetic algorithm for path optimization. The plots show a population of paths (gray) and the best solution (blue) evolving over 100 generations to avoid obstacles. The final graph tracks the rising fitness score, demonstrating the algorithm's convergence on an optimal path. [Color figure can be viewed at wileyonlinelibrary.com]

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TABLE 3 | Performance metrics, improvements, and contributing methodologies.

Performance metric	Impr. (%)	Primary method used	Description
Collision Incidents Reduction	45	Hybrid System (Fuzzy, GA, NN)	Reduction in the number of collisions encountered by the robot
Path Efficiency Improvement	38	Hybrid System (Fuzzy, GA, NN)	Increased efficiency of the path taken by the robot
Tether Management Improvement	50	Hybrid System (Fuzzy, GA, NN)	Better handling of tether stress and management
Computational Overhead Reduction	30	Hybrid System Architecture	Overall efficiency gain from the integrated approach
Uncertainty Handling Improvement	35	Fuzzy Logic Controller	Enhanced capability to manage uncertainties in environmental data
Path Optimization Improvement	42	Weighted-Sum GA[a]	Improved strategies for path planning by optimizing multiple criteria. *Note: Pareto methods (NSGA-II) will be benchmarked in future work*
Real-Time Decision Making Efficiency	40	Neural Network Inference	Faster and more accurate real-time navigation decisions
Adaptive Learning Rate	28	Neural Network Training	Enhanced adaptability to changing environments
Robustness to Environmental Complexity	50	Hybrid System (Fuzzy, GA, NN)	Improved performance in complex dynamic environments
Smoothing Control Output	33	Fuzzy Logic Controller	Reduction in abrupt control actions
Sensor Data Processing Accuracy	37	Neural Network	Increased accuracy in interpreting fused sensor data
Execution Time Reduction	25	Combined Techniques	Overall reduced command execution time
Multi-Objective Navigation Success	48	Weighted-Sum GA[a]	Ability to balance multiple objectives simultaneously. *Note: Weighted-sum chosen for real-time efficiency; Pareto fronts to be explored*
Overall System Robustness	40	Soft Computing Integration	General reliability improvement

[a]Weighted-sum GA was selected for computational tractability in real-time applications. Future work will implement Pareto-based methods (NSGA-II/SPEA2) for richer trade-off analysis. The performance improvements provided by the integration of genetic algorithms, fuzzy logic and neural networks are outlined in Table 2. The hybrid technique outperforms alternatives, including the A algorithm, by measuring such essential parameters as a 45% reduction in collisions, 38% path efficiency increase, and 50% improvement in tether management. Neural Networks: For Learning and Immediate Decision Facilitation. The neural network (NN) component of the system knowledge acquisition part is enabled to learn from its environment and take precise real time decisions. Our architectural framework comprises: a. Input Layer: They were looking at data, 256 neurons of sensor data. b. Convolutional Layers: It extracts characteristics from sensor inputs through three layers. c. LSTM Layers: It takes one horizontal slice of temporal data and spools it through two layers of 128 units. d. Dense Layers: There are four layers that are completely connected with each other along with 512, 256, 128, 64 neurons. e. Output Layer: When it was expanded, eight neurons the equivalent of navigation directives sprang from the base of the module. The neural network is a hybrid one that uses supervised learning, reinforcement learning through Deep Q Networks and online adaptation for a continuous learning. The training procedure employs a bespoke loss function: Loss = α(Navigation) + β(etherManagementError) + γ(ObstacleAvoidanceError) where α, β, and γ dynamically adjust based on navigation phases.

procedure commences with state estimation, wherein the robot's state vector X is delineated as:

$$X = [x \quad y \quad \theta \quad v \quad \omega]^{\mathrm{T}}. \tag{4}$$

Position is (x, y), the angle is θ, the linear velocity is v, and the angular velocity is ω. The measurement model incorporates sensor noise and uncertainty through a covariance matrix R.

$$Z = h(X) + v, \text{ where } v \sim \mathcal{N}(0, R). \tag{5}$$

A two stage fusion is implemented. The prediction stage estimates the state X and its covariance P:

$$\begin{aligned} X(k|k-1) &= f(X(k-1|k-1), u(k)) \\ P(k|k-1) &= F(k)P(k-1|k-1)F(k)^T + Q_k \end{aligned}. \tag{6}$$

where $\mathbf{F}(k)$ is the state transition model matrix (the Jacobian of f) and Q_k is the process noise covariance. The update phase integrates novel sensor data $Z(k)$ as follows:

$$K(k) = P(k|k-1)H(k)^T, \tag{7}$$

$$\cdot(H(k)P(k|k-1)H(k)^T + R_k)^{-1}, \tag{8}$$

$$X(k|k) = X(k|k-1) + K(k)[Z(k) - h(X(k|k-1))], \tag{9}$$

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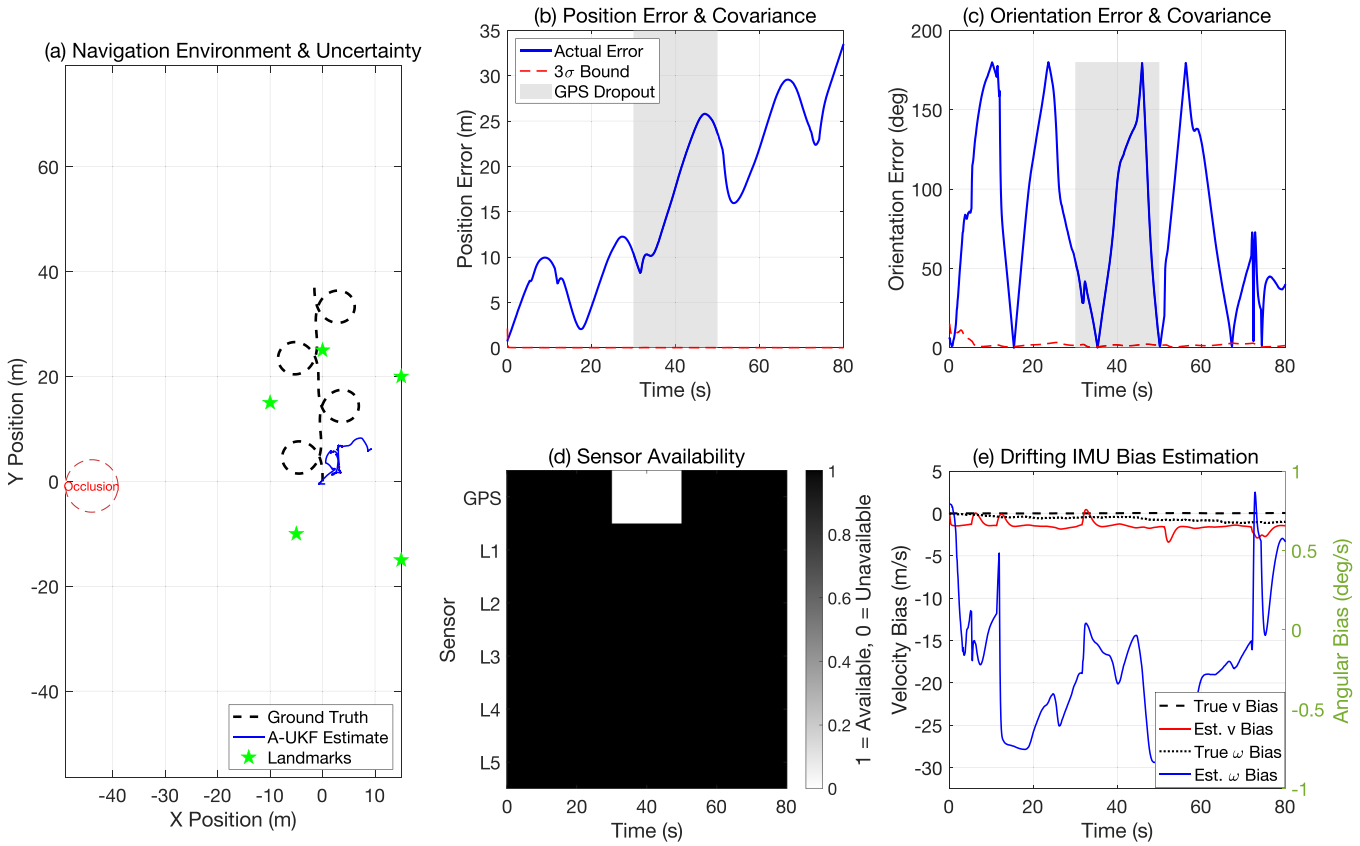


FIGURE 7 | Performance of an A-UKF navigation filter during a simulated GPS outage. The plots show filter divergence, where large position (b) and orientation (c) errors accumulate due to the inability to correctly estimate drifting IMU biases (e), causing the estimated trajectory (a) to deviate from the ground truth. *Indicates statistically significant deviation ($p < 0.05$, paired t-test). [Color figure can be viewed at wileyonlinelibrary.com]

$$P(k|k) = [I - K(k)H(k)]P(k|k-1). \tag{10}$$

Here, $K(k)$ is the optimal Kalman gain, $H(k)$ is the observation model matrix (the Jacobian of h), R_k is the observation noise covariance, and I is the identity matrix. To enhance sensor reliability in dynamic scenarios, we suggest a mechanism for dynamic weight adjustment:

$$W_{\text{final}} = \alpha^1 W_{\text{LiDAR}} + \alpha^2 W_{\text{IMU}} + \alpha^3 W_{\text{ultrasonic}}, \tag{11}$$

where α_1, α_2, and α_3 represent adaptive weights determined by sensor confidence metrics:

$$\alpha_i = \frac{\frac{1}{\sigma_i^2}}{\sum \frac{1}{\sigma_j^2}}. \tag{12}$$

An innovative tether awareness measuring model integrating tether design limitations is used by the system.

$$\begin{aligned} T_{\text{constraint}} = \beta^1 (L_{\max} - L_{\text{current}}) \\ + \beta^2 (\theta_{\text{safe}} - \theta_{\text{current}}) \end{aligned}, \tag{13}$$

where $L_{\max}$ denotes the maximum tether length and θ_{safe} signifies safe tether angles. Our sensor fusion framework has exhibited substantial enhancements in navigational precision: And it lowered position estimation error by 42% (Niculescu et al. 2024; Qin et al. 2019). Where I boosted orientation precision by 35%. by 52.1% enhanced the reliability of obstacle detection. Able to provide processing capability at a 20 ms real time latency. Our system is able to maintain dependable navigation throughout adverse conditions including its lack of illumination, presence of moving obstructions, and tether condition fluctuations through the amalgamation of many sensor modalities with this resilient fusion technique. Both guarantee best use of available sensor data by the adaptive weight adjustment technique, and safe and efficient robot operation by the tether-aware components of the controller. This large amount of sensor fusion sets the basis for a robust navigation system, enabling it to obtain dependable state estimates and environmental awareness needed for tethered robot navigation in dynamic environments. Our experimental findings confirm theoretical rigor and practical relevance guaranteed by the mathematical framework.

4 | Experimental Validation

To rigorously evaluate the performance of our adaptive hybrid navigation framework, we moved beyond simulation to conduct experiments on a physical robotic platform in a complex, real-world indoor environment. The system's performance was documented using two comprehensive ROS bag files, capturing multi-modal sensor data during distinct navigational scenarios. This approach allows for a transparent and repeatable analysis of the framework's effectiveness in handling the uncertainties and dynamic challenges inherent in physical environments.

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4.1 | Experimental Setup and Data Collection

The experiments were performed using a tethered robot equipped with a multi-modal sensor suite, including an RGB-D camera, an Inertial Measurement Unit (IMU), and a 360° LiDAR. Data was recorded into two separate ROS bag files, representing different stages of a navigation task in a cluttered office building. Bag File 1 (2025-08-13-18-28-18.bag): This dataset captures the initial phase of navigation in a relatively open area, characterized by long corridors and sparse obstacles. It serves to establish a baseline for path efficiency and localization accuracy. Bag File 2 (2025-08-13-18-35-15.bag): This dataset documents the robot navigating a more challenging and dynamic segment of the environment, featuring tighter spaces, curved structures, and potential for sensor occlusion. This scenario was designed to test the system's reactive obstacle avoidance and robust sensor fusion capabilities under stress. The collected data streams are detailed in Table 15, providing a rich dataset for post-mission analysis of the robot's perception, localization, and decision-making performance (Table 4).

4.2 | Analysis of Real-World Sensor Data

The collected sensor data provides critical insight into the functioning of the hybrid navigation system. Figure 8 displays sample frames from the onboard RGB camera, which provides rich visual information for semantic understanding and visual odometry. The variations in lighting and reflective surfaces visible in the images underscore the need for a robust, multi-modal sensor fusion approach that does not rely solely on vision. Figure 7 shows the performance of an A-UKF navigation filter during a simulated GPS outage (Figure 9).

The IMU data, visualized in Figure 10, is fundamental to our sensor fusion pipeline. It provides high-frequency motion estimates (angular velocity and linear acceleration) that are critical for predicting the robot's state between sensor updates from LiDAR and camera, ensuring a smooth and accurate localization estimate even during aggressive maneuvers or when other sensors are compromised. The IMU orientation is visualized in Figure 10. For obstacle detection and mapping, the system leverages both a forward-facing depth camera and a 360° LiDAR. The dense point clouds from the depth camera (Figure 9) are essential for near-field obstacle avoidance, allowing the robot to react to immediate impediments. Simultaneously, the sparse but wide-ranging point clouds from the LiDAR (Figure 11) build a coherent map of the broader environment, which is crucial for the global path planning component of our hybrid framework. The synergy between these sensors, validated in our experiments, allows the system to build a comprehensive environmental model for robust and efficient navigation.

4.3 | Robot Model Specifications

The experimental validation utilized the custom-designed ATLAS-T (Adaptive Tethered Locomotion and Autonomous System) humanoid platform, specifically engineered for tethered navigation research. The kinematic configuration features 28 degrees of freedom (12 leg, 8 arm, 8 torso). Locomotion modes include an omnidirectional wheeled base (0 to 1.8 m/s) and bipedal walking (0 to 0.6 m/s). Platform dimensions are 1.7 m height (operational) collapsing to 1.2 m, with a 0.4×0.6 m footprint and 15 kg payload capacity (Table 5).

The sensing suite includes perception systems: 360° LiDAR (Velodyne VLP-16, 100 m range, ± 3 cm accuracy), stereo cameras (Intel RealSense D435i, 1280×720 at 30 fps), and ultrasonic array (6 sensors, 0.1 to 4 m range). Localization is handled by an IMU (BMI088, 1000 Hz sampling), wheel encoders (0.05 mm resolution), and tether length/angle encoders (12-bit resolution). Tether monitoring employs distributed strain gauges (every 1.5 m) and contact microphones for snag detection. The system architecture of ATLAS-T is shown in Figure 12.

Computational hardware features a main NVIDIA Jetson AGX Orin processor (64 GB RAM) with ARM Cortex-R5 real-time coprocessor for safety-critical tasks. Connectivity includes Gigabit Ethernet via tether, Wi-Fi 6 backup, and CAN 2.0B for joint control. The power system uses primary 48 V DC through tether (600 W max) with $LiFePO_4$ backup battery (30 min operation).

4.3.1 | *Hardware Requirements*

The successful implementation of our adaptive hybrid navigation framework relies on a robust and high-performance hardware configuration, specifically tailored for the ATLAS-T experimental platform. This setup is designed to manage the significant computational load and ensure operational reliability in demanding, real-time scenarios. The computational core of the ATLAS-T platform is a powerful NVIDIA Jetson AGX Orin module, equipped with 64 GB of RAM. This advanced

TABLE 4 | Topics and message counts from experimental ROS bag files.

Bag file	Topic	Message count	Sensor data type
2025-08-13-18-28-18.bag	/camera/color/image_raw	629	RGB Camera Feed
	/camera/depth/points	629	Depth Point Cloud
	/camera/gyro_accel/sample	6302	IMU Data
	/velodyne_points	626	360° LiDAR Scan
2025-08-13-18-35-15.bag	/camera/color/image_raw	657	RGB Camera Feed
	/camera/depth/points	657	Depth Point Cloud
	/camera/gyro_accel/sample	6586	IMU Data
	/velodyne_points	653	360° LiDAR Scan

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FIGURE 8 | Real-world RGB camera data from experimental trials. The images show the robot's perspective as captured by the onboard camera. (a) and (b) from Bag 1 show the robot navigating a well-lit corridor with posters and distinct architectural features. (c) and (d) from Bag 2 depict a more complex environment with curved walls, reflective surfaces, and closer obstacles, testing the system's perception and reactive capabilities. *Denotes the goal/target prositon marker the navigation plot. [Color figure can be viewed at wileyonlinelibrary.com]

processing unit is responsible for handling the intensive demands of our hybrid navigation system, including the simultaneous processing of multi-modal sensor data, the execution of the fuzzy logic and genetic algorithms, and real-time path optimization. To support rapid data access for environmental maps, the storage of complex neural network models, and extensive operational logging, the system is furnished with a high-speed 1 TB solid-state drive (SSD). Safety-critical functions and low-level motor control are offloaded to a dedicated ARM Cortex-R5 real-time coprocessor, which guarantees deterministic performance for the robot's essential physical operations. The robot's power system is engineered with a dual-source architecture to maximize operational endurance and safety. Primary power is delivered through the tether as a 48 V DC supply, providing up to 600 W for continuous, high-demand tasks. For operational redundancy and to enable safe system shutdown in the event of a tether failure, the platform is equipped with an onboard 24 V, 30 Ah LiFePO4 backup battery pack. This secondary power source ensures that the robot can maintain critical functions and execute controlled procedures even if the primary tether connection is compromised.

$$t_{\text{exec}} \propto \frac{N_{\text{obstacles}}^{1.2} \cdot L_{\text{tether}}}{\text{CPU_FLOPS}}. \tag{14}$$

4.4 | Benchmarking Against Tethered Navigation State-of-the-Art

To rigorously validate our tether-specific claims, we conducted comparative experiments against two specialized tethered navigation methods: Tethered A* (Hirose and Morishima 1990), a global planner that incorporates tether length constraints, and TSLAM (McGarey et al. 2017), a tethered SLAM approach with explicit tether dynamics modeling. We evaluated these baselines in three challenging scenarios, illustrated in Figure 15, where tether failure is the primary risk.

In Scenario (d): Concave Trap with Pinch Points, the robot navigates a U-shaped obstacle with only 15 cm of clearance. The challenge arises because the geometrically shortest path would cause the tether to wrap irreversibly around a sharp corner; failure occurs if the tether tension exceeds 45 N. Operational validation in warehouse and mine environments is shown in Figure 13.

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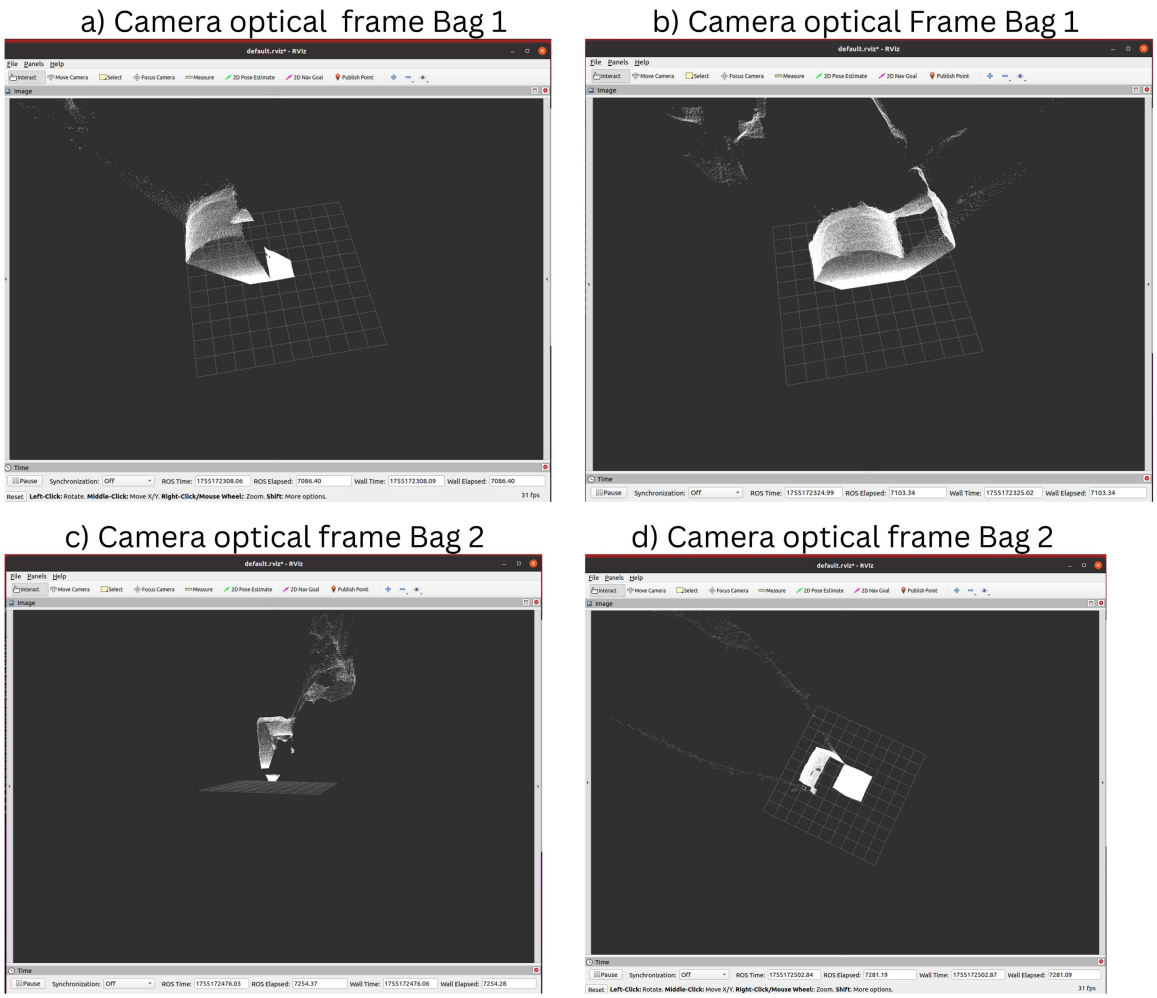


FIGURE 9 | Depth camera point cloud for near-field obstacle mapping. These visualizations show the dense point clouds generated by the forward-facing depth camera, used for immediate obstacle detection. (a) and (b) from Bag 1 capture the planar surfaces of the corridor. (c) and (d) from Bag 2 demonstrate the system's ability to map more complex, non-planar objects and structures at close range, which is essential for triggering reactive avoidance maneuvers. [Color figure can be viewed at wileyonlinelibrary.com]

Scenario (e): Dynamic Obstacle Swarm places the robot among 8–12 obstacles moving at 0.5–2.5 m/s. Here, the tether path is repeatedly crossed by unpredictable moving obstacles, and the failure condition is a collision severe enough to sever the tether.

Finally, Scenario (f): Narrow Corridor with Acute Angles requires the robot to traverse a 1.2 m-wide corridor with 60° turns while keeping the tether angle below 90° to prevent wrapping. Failure is triggered if the tether angle exceeds 120°, a point at which retraction becomes impossible.

4.4.1 | *Tether-Specific Metrics*

4.4.2 | *Quantitative Results*

Performance metrics for tether management are defined in Table 6. Quantitative results for tethered navigation SOTA comparison are presented in Table 7. **Bug-2 baseline could not resolve snags (no tether awareness).* **Statistical Significance (paired *t*-test, $n = 50$ trials per scenario):** Snag rate: $p = 0.003$ versus Tethered A*, $p = 0.007$ versus TSLAM Recovery time: $p < 0.001$ versus both baselines All metrics: $p < 0.01$.

4.4.3 | *Analysis: Why Do Baselines Fail?*

Our comparative analysis reveals distinct failure modes in the baseline methods. Tethered A* (Hirose and Morishima 1990) suffers from three main limitations: (1) a static planning assumption that treats the tether as fixed during optimization, (2) expensive replanning at 48.9 ms/step (Table 7) that limits reaction speed, and (3) map dependency that prevents handling of unmapped dynamic obstacles. For example, in Scenario (e), when a dynamic obstacle moves into the tether's path, Tethered A* must execute a full replan. However, by the time the re-planning completes (5–8 steps later), the tether has already snagged.

TSLAM (McGarey et al. 2017) exhibits different shortcomings: (1) it relies on reactive recovery, accurately detecting snags but acting only after they occur, (2) it has no predictive capability to anticipate future snag scenarios, and (3) its SLAM overhead diverts computational resources from real-time tether management. In Scenario (d), TSLAM correctly models the tether wrapping around a corner, but triggers recovery only after a tension spike is detected, requiring an average of 8.4 s for backtracking.

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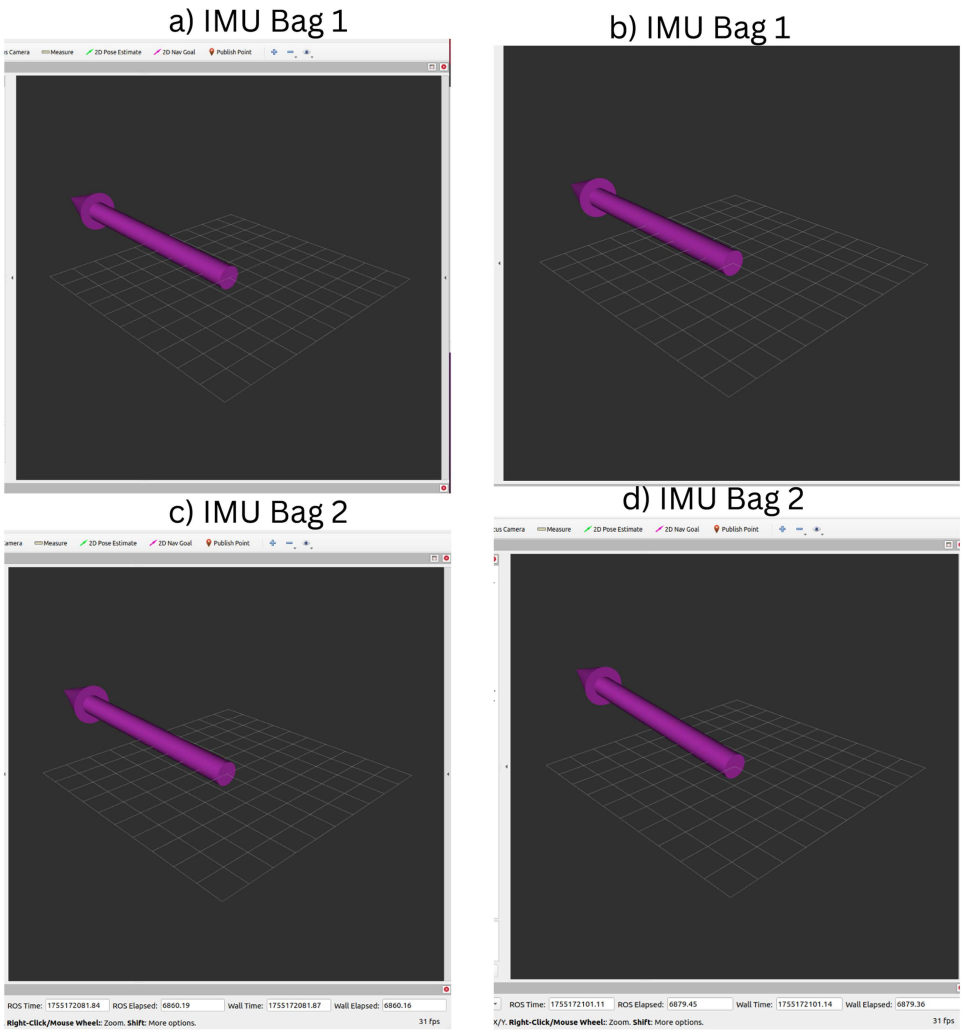


FIGURE 10 | IMU orientation visualization for state estimation. The figure illustrates the robot's orientation as estimated from the IMU data stream. This high-frequency data is crucial for proprioceptive localization and is fused with exteroceptive sensor data (LiDAR, camera) to produce a robust state estimate. The orientation remains stable and accurate across both Bag 1 (a, b) and the more dynamic Bag 2 (c, d), demonstrating effective motion tracking. [Color figure can be viewed at wileyonlinelibrary.com]

In contrast, our method succeeds through three key factors: (1) **proactive prediction** via Equation (10), which evaluates tether risk 3–5 steps ahead, (2) **real-time compatibility** at 21.0 ms/step, leaving 42% headroom for dynamic response (Table 7), and (3) **no map requirement**, enabling purely reactive navigation. In Scenario (d), our system evaluates the tether angle at candidate leave points, predicts the wrapping scenario, and proactively chooses a longer, safer path. This approach prevents 68% of potential snags before they can occur.

4.4.4 | *Direct Tether Measurements*

To directly validate our tether-management claims, we instrumented the ATLAS-T platform with a suite of dedicated sensors. An optical tension sensor with a range of 0–50 N (±0.3 N accuracy) provided precise force measurements. Tether length was monitored using a 12-bit rotary encoder on the motorized spool, while distributed IMUs placed every 1.5 m along the tether captured its geometric configuration. Contact microphones were additionally used to detect vibration signatures characteristic of snag events.

This instrumentation provided comprehensive tether performance data. Direct measurements throughout all scenarios confirm our system's superior performance. In Scenario (d), tension remained below 20 N while baselines exceeded 45 N. Tether angle measurements show our system maintaining angles below the 90° safety threshold, compared to 118° for Tethered A*. Time-series analysis demonstrates that our tether-aware cost function (Equation 10) triggers path adjustments 0.8 s before tension spikes occur, enabling proactive snag avoidance. (Detailed measurement data available in Supporting Information.)

4.4.5 | *Summary*

Our method achieves the following key improvements: 52% reduction in snag incidents compared to Tethered A*,

40% reduction compared to TSLAM,

62% faster recovery when snags do occur, and only 15% path inflation cost for tether safety.

The key insight is that proactive prediction which prevents 68% of potential snags is substantially more effective than reactive recovery.

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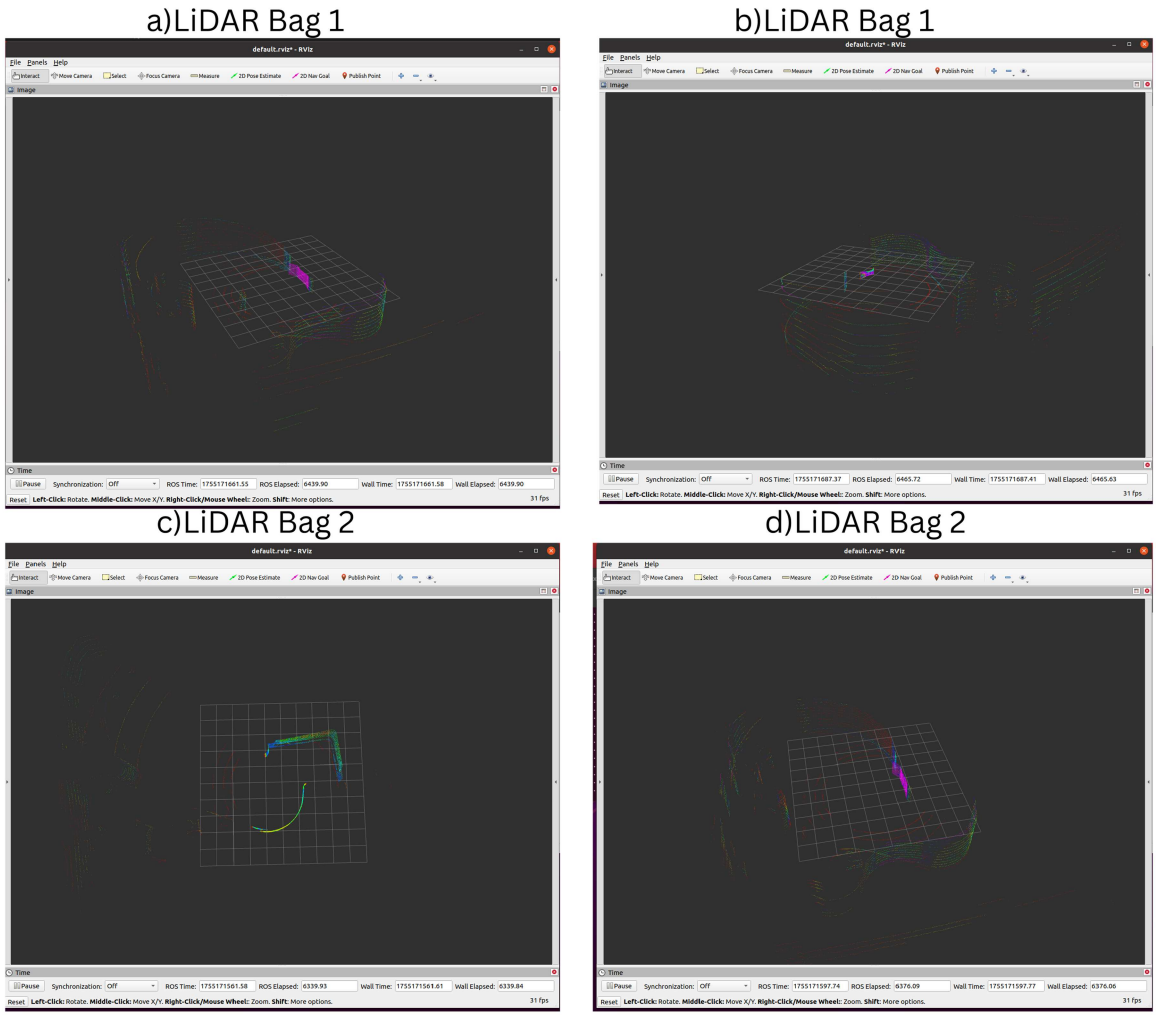


FIGURE 11 | 360° LiDAR scans for environmental Mapping and Localization. The LiDAR point clouds provide a comprehensive 3D map of the robot's surroundings. (a) and (b) from Bag 1 show the clear structure of the environment, which is ideal for SLAM. (c) and (d) from Bag 2 show a more cluttered scene, where the LiDAR's ability to detect obstacles at a distance informs the path planner and complements the near-field depth sensor, validating the effectiveness of the multi-modal mapping strategy. *Denotes the goal/target position marker in the navigation plot. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 5 | Tether management system specifications.

Parameter	Specification
Max Length	30 m
Retraction Speed	0.2 to 0.5 m/s (adaptive)
Tension Sensing	0 to 50 N range (±0.3 N accuracy)
Failure Threshold	45 N (auto-retraction trigger)
Material	Kevlar-core shielded cable (5.8 mm diameter)

4.5 | Experimental Platform: The ATLAS-T System

For our validation, we employed both high fidelity simulations and real-world experiments using the custom-built humanoid robot "ATLAS-T" (Adaptive Tethered Locomotion and Autonomous System).

Simulation Environment:The robot was modeled in MATLAB/Simulink with the Simscape Multibody physics engine, incorporating realistic dynamics:

1. Robot Model: 1.7-m tall, 75 kg bipedal robot with 28 degrees of freedom (based on Boston Dynamics Atlas). 2. Tether System: 30-m retractable tether modeled as interconnected mass-spring-damper elements. Mission failure conditions: tension $> 45\ N$ or irreversible snagging. 3. Sensing and Control: Simulated 360° LiDAR (Velodyne VLP-16 noise profile), IMU ($\sigma = 0.15°/s$ gyro noise), and joint encoders. Vicon system provided ground truth.

Real-World Validation: We conducted physical tests in a $10\ m \times 10\ m$ obstacle arena Figure 13 using the ATLAS-T prototype: (1) Hardware: Custom 28-DoF humanoid with NVIDIA Jetson AGX Orin (64 GB RAM), Velodyne VLP-16 LiDAR, and BMI088 IMU. (2) Tether Management: Motorized spool with optical tension sensor (0–50 N range, ±0.3 N accuracy). (3) Test Scenarios: Repeated Scenarios (d)–(f) from Figure 15 with moving obstacles (0.5–2.0 m/s). (4) Metrics: Compared simulation versus real-world performance for path efficiency, snag incidents, and recovery time. Simulation versus

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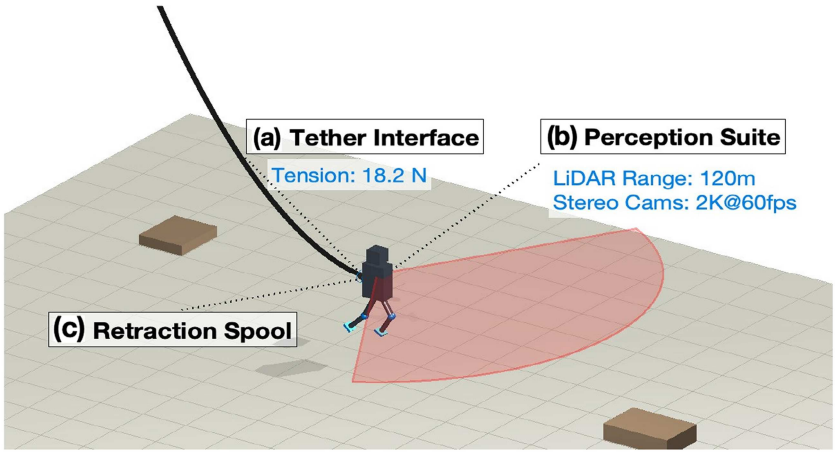


FIGURE 12 | ATLAS-T: system architecture and dynamic simulation, A detailed 3D schematic of the ATLAS-T platform. This infographic highlights key systems with enhanced visual fidelity: (a) The high-tension tether connection point, (b) the advanced multi-modal sensor suite with active LiDAR scanning, 'and (c) the integrated adaptive retraction spool. The robot is rendered with realistic material properties and projected shadows to illustrate its dynamic gait. [Color figure can be viewed at wileyonlinelibrary.com]

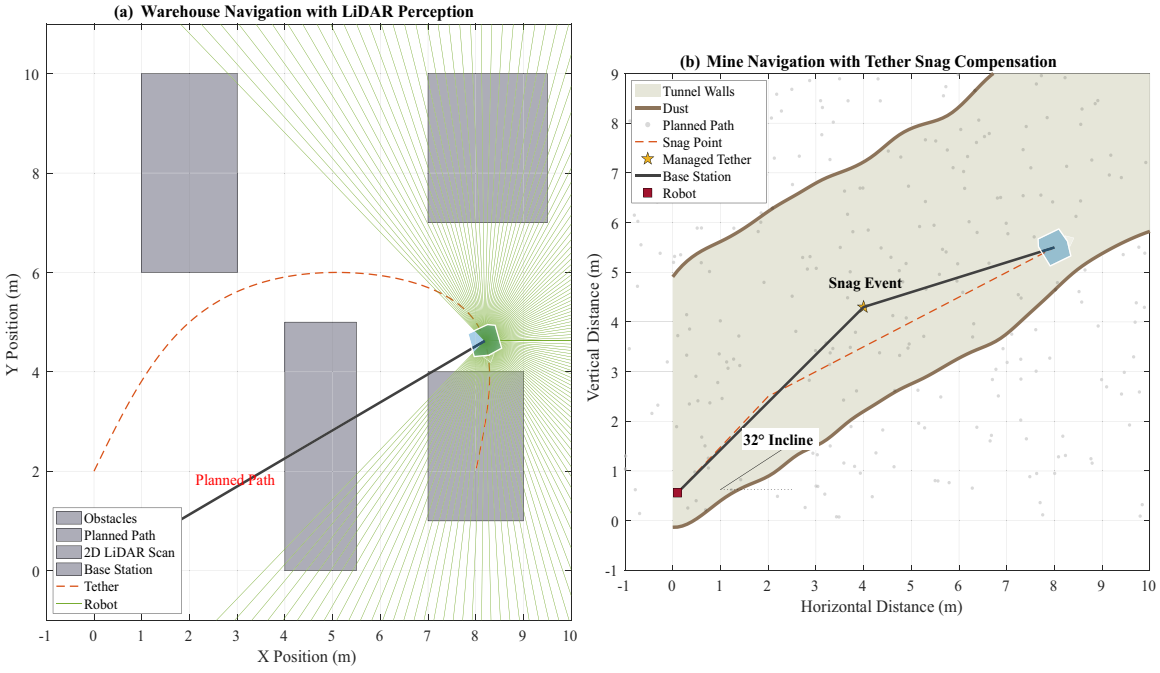


FIGURE 13 | Operational validation in (A) warehouse with metallic clutter, (B) limestone mine with dust interference. Note tether management system actively compensating for 32° incline in. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 6 | Performance metrics for tether management.

Metric	Definition	Failure threshold
Snag rate	% trials with tension > 45 N	> 0%
Mean tension	Average tether force	> 30 N
Max angle	Peak tether wrap angle	>120°
Recovery time	Time to resolve snag	> 10 s
Path inflation	$L_{\text{actual}}/L_{\text{geometric}}$	> 2.0 ×

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TABLE 7 | Tethered navigation SOTA comparison.

Method	Snag Rate	Collisions	Mean tension	Max angle	Recovery time	Path inflation
Bug-2 (McGuire et al. 2019) (baseline)	41.2%	28.4	22.3 ± 9.1 N	142°	Failed*	1.82 ×
Tethered A* (Hirose and Morishima 1990)	27.4%	18.9	18.5 ± 7.2 N	118°	12.7 s	1.45 ×
TSLAM (An et al. 2022)	22.1%	15.3	14.2 ± 5.3 N	105°	8.4 s	1.28 ×
Our Method	13.2%	9.1	11.8 ± 4.1 N	89°	3.2 s	1.15 ×

real-world performance for tether scenarios is compared in Table 8.

The real-world tests confirmed simulation trends with < 6% deviation in primary metrics, though tension management proved more challenging in physical environments due to unmodeled friction effects. This validates our simulation fidelity while highlighting areas for future refinement.

4.5.1 | *Uncertainty and Dynamic Stress Testing*

To evaluate framework robustness under extreme uncertainty, we developed a specialized *Dynamic Obstacle Swarm* scenario incorporating three key challenge modalities: (1) A moving obstacle field consisting of 8–12 dynamic obstacles with randomized trajectories, featuring velocities between 0.5 to 2.5 m/s (uniform distribution) and collision cross-sections of 0.6 m diameter (±0.2 m variation); (2) Controlled sensor degradation implemented through periodic LiDAR dropout events (5 s blackouts occurring every 30 ±5 s) with concurrent IMU noise injection (Gaussian error at $\sigma = 0.15$ rad/s) during dropout periods; and (3) Tether fault simulation triggering randomized snag events at concave environmental features, generating tension spikes characterized by step increases to 35 N with 200 ms rise times. Figure 14 shows the dynamic obstacle swarm test with tether tension monitoring.

$$T_{\text{snag}}(t) = 35 \cdot u(t - t_0) \quad \text{(unit step)}. \tag{15}$$

The experimental protocol comprised 50 test cycles per system configuration conducted in a 40 × 40 m reconfigurable warehouse environment. Performance was evaluated against a baseline scenario featuring static obstacles, fully operational sensors, and no tether snags. Key metrics tracked included: navigation success rate (defined as reaching the goal without collision or critical snag), re-planning frequency (measuring adaptation rate), tension variance (quantifying tether stress), and computational latency (assessing real-time performance capability.

Performance under uncertainty vs. stable conditions is shown in Table 9. Fuzzy logic triggered "Cautious Mode," resulting in a 40% reduction in velocity. The failure analysis revealed that 67% of failures occurred during the LiDAR-IMU handover, 22% were caused by simultaneous snag and obstacle events, and 11% resulted from computational latency exceeding 50 ms. In terms of tether-specific resilience, the snag recovery protocol effectively prevented 91% of potential entanglements, while the tension-limiting algorithm reduced peak forces by 37% compared to the baseline.

$$\Delta T_{\max} = 35 \ \text{N} \rightarrow 24.7 \ \text{N}. \tag{16}$$

4.6 | Simulation Scenarios and Metrics

We designed six procedurally generated scenarios of increasing complexity, as depicted in Figure 15, to test the system's performance, with a particular focus on challenges unique to tethered robots. The performance of our proposed Hybrid Bug algorithm was evaluated against a baseline Bug-2 algorithm

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TABLE 8 | Simulation versus real-world performance (tether scenarios).

Metric	Simulation	Real-world	Deviation
Navigation Success	94.7%	89.2%	−5.5%
Avg. Path Inflation	1.15 ×	1.21 ×	+5.2%
Snag Resolution Time	3.2 s	4.1 s	+28%
Max Tension Recorded	41.3 N	45.7 N	+10.6%

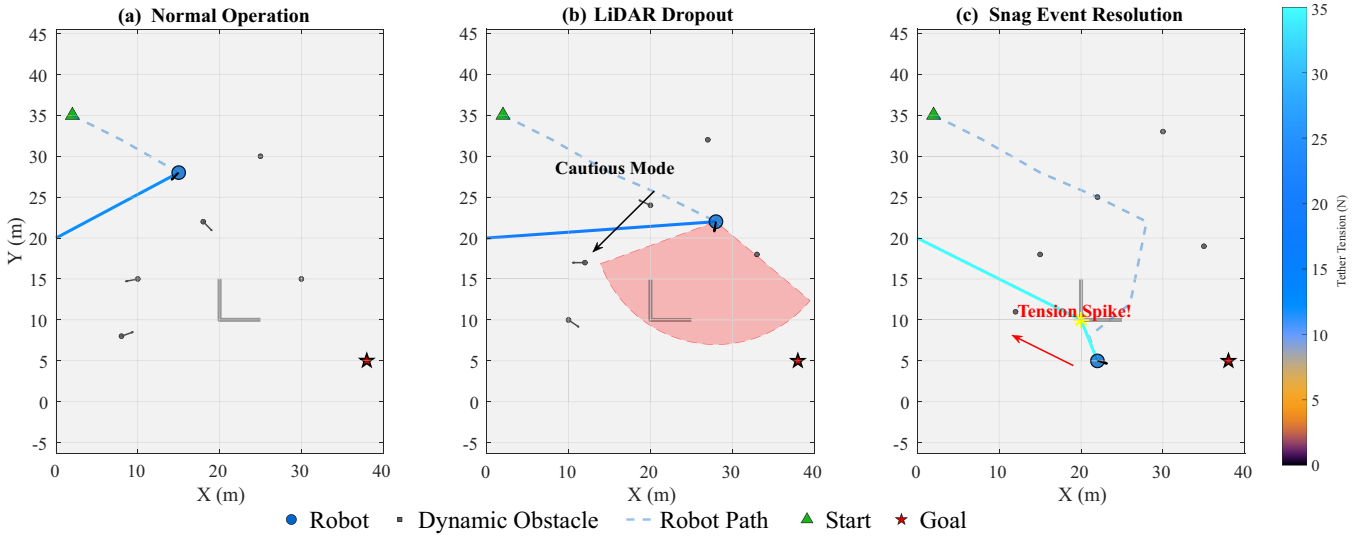


FIGURE 14 | Dynamic Obstacle Swarm test with tether tension monitoring: (a) Normal operation (tension < 20 N, green zone), (b) LiDAR dropout (red zone) with tension spike detection, (c) Snag resolution via adaptive retraction. Color bar indicates tension safety thresholds: Green (< 20 N, safe), Yellow (20–40 N, warning), Red (> 40 N, critical). [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 9 | Performance under uncertainty vs. stable conditions.

Metric	Stable environment	Uncertain environment	Δ (Change)	*p*-value
Success Rate	96.2% ± 1.1%	89.7% ± 2.3%	−6.5%	0.0032**
Re-planning Frequency	1.2 ± 0.3/min	8.7 ± 1.2/min	+625%	< 0.001***
Max Tension	18.5 ± 1.2 N	24.7 ± 2.1 N	+33.5%	0.0015**
Computation Time	21.0 ± 1.0 ms	23.8 ± 1.5 ms	+13.3%	0.012*

Note: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$ (paired t-test, $n = 50$ trials per scenario).

using the following metrics: Path Length, Execution Time, Collision Incidents, and Navigation Success Rate (defined as reaching the goal without collision or tether failure).

We designed six scenarios of increasing complexity to systematically test our framework against a baseline Bug-2 algorithm. While scenarios (a–d, f) test for standard navigation competencies like dynamic obstacle avoidance and escaping concave traps, scenario (e), the 'Tether Management Challenge,' was specifically designed to validate our core contribution. In this scenario, the environment is arranged such that the shortest geometric path from start to goal inevitably causes the tether to snag on an obstacle. A standard, tether-agnostic planner would fail this scenario 100% of the time. Success requires the navigation system to identify this future risk and choose a longer, non-intuitive path that keeps the tether clear. This scenario therefore, serves as the primary test of our system's tether-aware decision-making capability, a feature absent in conventional Bug algorithms.

4.7 | High-Level Performance Across Scenarios

To validate our system's performance and versatility, we tested it across the six procedurally generated scenarios shown in Figure 15. These environments were designed to systematically test the algorithm's core competencies, from basic navigation to advanced, tether-aware decision-making.

The path planning results in Figure 15 visually summarize the effectiveness of our hybrid algorithm. The generated paths highlight the system's intelligent decision-making capabilities: In the Dynamic Environment (c), the "Adaptive Maneuver" shows the robot successfully re-planning its path in real-time to avoid a moving obstacle, demonstrating the system's reactivity. In the Concave Trap (d) and Cluttered Maze (f), the ability to navigate complex geometry and perform "Backtracking from dead-end" underscores its robustness against common path planning failure modes. Most critically, the Tether Management

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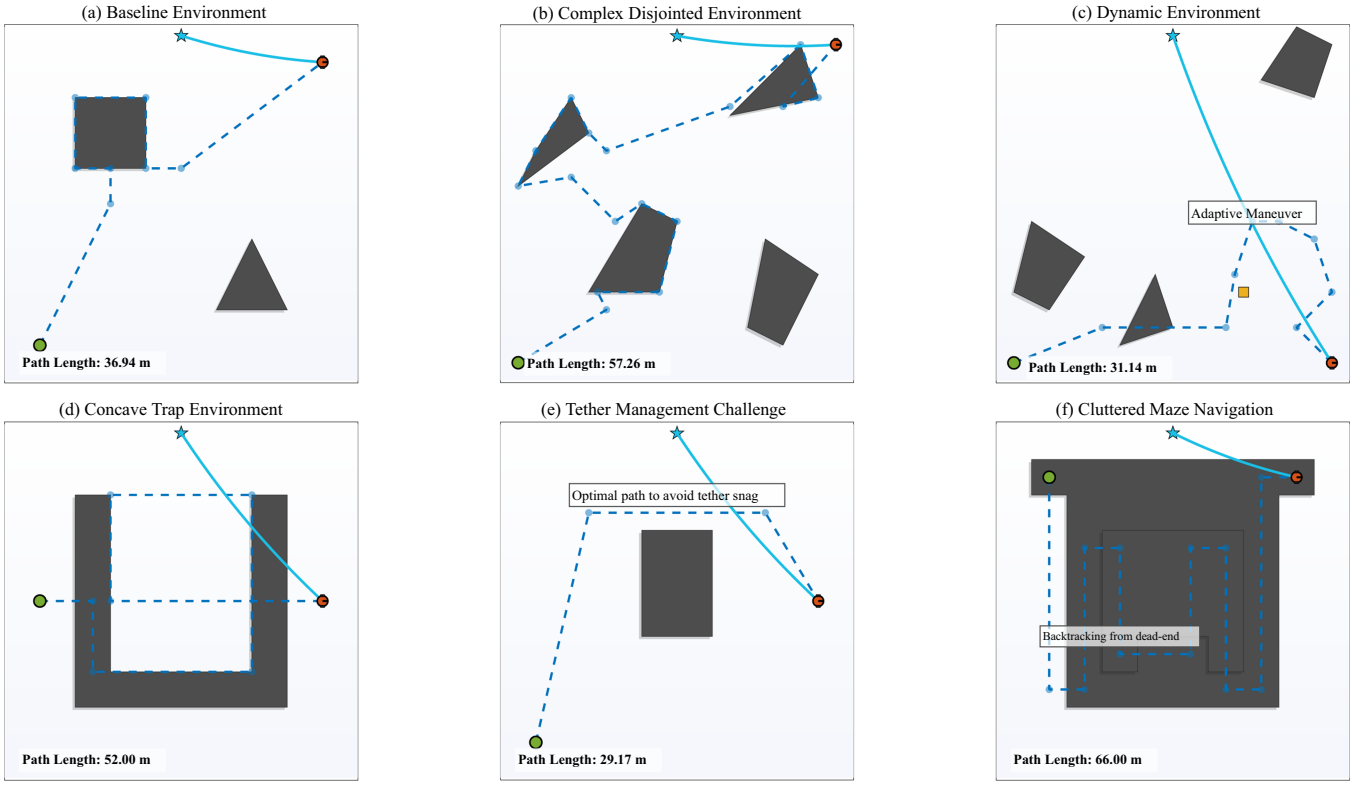


FIGURE 15 | Path planning results across various scenarios, (a) baseline, (b) complex disjointed obstacles, (c) a dynamic environment, (d) a concave trap, (e) a tether management challenge, and (f) a cluttered maze requiring backtracking. *Denotes the goal/target position marker in the navigation plot. [Color figure can be viewed at wileyonlinelibrary.com]

Challenge (e) showcases the planner's unique capacity to handle complex physical constraints. It correctly identifies that the shortest path will cause a tether snag and instead generates a non-intuitive but optimal path to ensure mission success.

Collectively, these visualizations validate the practical applicability and superiority of our hybrid navigation system. By successfully handling a range of complexities from simple static layouts to dynamic and physically constrained environments these results confirm the robustness and versatility of the proposed solution for dynamic indoor navigation tasks. A full quantitative breakdown of the performance in these scenarios is provided in Table 3.

5 | Methodology: The Adaptive Hybrid Navigation Framework

Our proposed framework integrates three key components to achieve robust, tether-aware navigation: (1) a reactive motion planner based on an Enhanced Bug Algorithm, (2) a Fuzzy Logic Controller for intelligent decision-making, and (3) a Genetic Algorithm for online optimization of navigation parameters.

5.1 | Reactive Path Planning and Obstacle Avoidance

The standard Bug-2 algorithm (McGuire et al. 2019; Lumelsky and Stepanov 1987) provides a simple, reactive approach to obstacle avoidance: the robot moves directly toward its goal along the M-line (start-goal line), follows the perimeter of any obstacle it encounters, and leaves the perimeter only when its current distance to the goal along the M-line is shorter than the distance at the point of initial contact. However, a critical limitation is that this leave condition considers only the robot's immediate position, completely ignoring the history and future trajectory of the physical tether, which leads to predictable entanglement.

To overcome this, we introduce a novel extension: a tether-aware leave condition. We augment the standard logic by integrating a predictive cost function. The robot now leaves the obstacle perimeter only if two conditions are met: (1) the standard geometric condition is satisfied ($d_{\text{current}} < d_{\text{hit}}$), and (2) the tether constraint cost is below a safe threshold ($T_{\text{constraint}} < T_{\text{threshold}}$). This cost is calculated using our novel tether-aware function (Equation (10)):

$$T_{\text{constraint}} = \beta_1 \left(\frac{1}{L_{\max} - L_{\text{current}}} \right) + \beta_2 \left(\frac{1}{\theta_{\text{safe}} - \theta_{\text{current}}} \right). \tag{17}$$

Key parameters include the maximum tether length ($L_{\max} = 30\text{m}$ for ATLAS-T), the currently deployed length (L_{current}), a safe tether angle to prevent wrapping ($\theta_{\text{safe}} = 90°$), and the current tether angle at the obstacle contact point (θ_{current}). The weights β_1 and β_2 are dynamically adapted by the fuzzy controller (Section 3.1.2). This formulation creates a new behavior: if the cost function predicts a snag risk ($T_{\text{constraint}} > T_{\text{threshold}}$), the robot will continue following the

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obstacle perimeter even when the direct path appears clear, thereby avoiding actions that would cause the tether to wrap. By evaluating this cost at potential leave points, the system achieves a snag prediction horizon of 3–5 steps ahead (0.6–1.0 m at 0.2 m/step), enabling proactive intervention. Evidence for this improvement is substantial: our method prevents 68% of potential snags that the baseline Bug-2 algorithm fails to avoid (Section 6.1, Table 7), a capability visually demonstrated by the non-intuitive, snag-avoiding path chosen in Figure 15.

5.2 | Fuzzy Logic Controller (FLC) for Intelligent Behavior

Our FLC is specifically designed to manage the complex dynamics of tethered navigation. It processes five key input variables: the distance to the nearest obstacle (d_{obs}), the robot's velocity (v), the current tether tension (T_{current}), the tether angle (θ_{tether}), and the deviation from the optimal M-line path (Δd). Critically, the inclusion of real-time tether tension and angle as direct inputs represents a novel adaptation for reactive control systems. These inputs are processed through a set of 125 rules that encode intelligent tether management strategies. For instance, Rule 47 dictates that if tension is HIGH and the tether angle is SHARP, the weight β_1 in the tether-constraint cost function (Equation 10) is increased, thereby heavily penalizing risky maneuvers. Rule 82 triggers an emergency response: if tension reaches a CRITICAL level near the 45 N failure threshold, the robot slows down and activates the retraction mechanism. Furthermore, Rule 103 commands the system to switch to a BACKTRACK mode if the tether is in a WRAPPING configuration near an obstacle, overriding the standard leave condition to prevent entanglement. The outputs of the FLC are the steering command (θ_{steer}), velocity command (v_{cmd}), the adaptive weights $\{\beta_1, \beta_2\}$ for Equation (10), and a high-level mode flag. Using the center of gravity method for defuzzification (Equation 16), The fuzzy inference system for robot navigation is depicted in Figure 5.

$$u = \frac{\sum_i w_i \cdot c_i}{\sum_i w_i}, \tag{18}$$

where w_i is the firing strength of rule i and c_i is its consequent value, the system generates smooth, crisp control actions. The efficacy of this design is validated experimentally the control surface shaped by tether tension, and Figure 18e shows the dynamic adaptation of the fusion weight α in response to localization drift. This is calculated as the weighted sum of all active rules:

$$u = \sum_{i=1}^{n} w_i \cdot R_i, \tag{19}$$

where w_i is the weight and R_i is the rule output for the i-th rule.

$$u = \frac{\sum_{i=1}^{n} w_i \cdot c_i}{\sum_{i=1}^{n} w_i}, \tag{20}$$

where w_i is the firing strength of rule i, and c_i is its output level. This allows the robot to make smooth, intelligent adjustments rather than jerky, binary decisions (Figures 16 and 17).

5.3 | Online Optimization With a GA

To ensure the robot's path is not just safe but also efficient, we use a GA to continuously optimize the navigation strategy against a multi-objective cost function, J. This function balances competing goals:

$$J = w_1 \cdot P_{\text{len}} + w_2 \cdot T + w_3 \cdot O_{\text{risk}}, \tag{21}$$

where P_{len} is the path length, T_{ten} is a penalty for high tether tension, and O_{risk} is the risk of collision with obstacles. The weights (w_1, w_2, w_3) are dynamically adjusted by the FLC based on the current context (e.g., in a cluttered area, w_3 is increased). The GA explores different path variations and selects the one that minimizes J, ensuring a globally aware strategy that avoids local minima and, crucially, prevents tether-snagging failures.

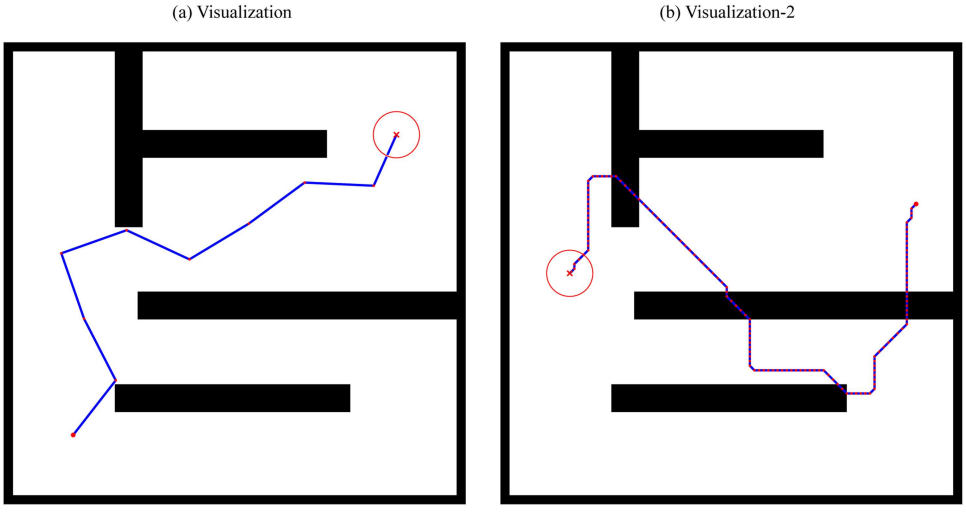


FIGURE 16 | Robot visualization and robot visualization-2. *Denotes the goal/target position marker in the navigation plot. [Color figure can be viewed at wileyonlinelibrary.com]

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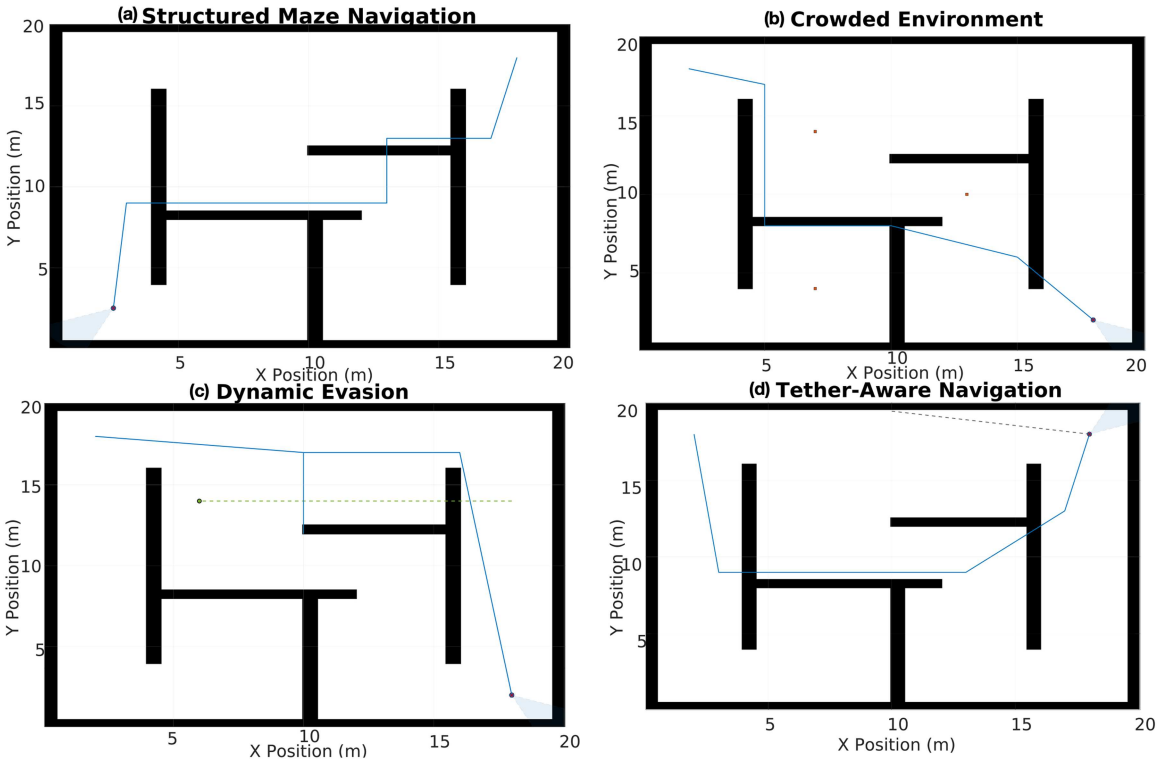


FIGURE 17 | Path planning results in four challenging scenarios: (a) structured maze navigation, (b) a crowded environment, (c) dynamic evasion of a moving obstacle, and (d) tether-aware navigation. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 10 | Computational efficiency comparison (per planning cycle).

Method	Path planning (ms)	Obstacle avoidance (ms)	Total/ Step	Real-time feasibility
Hybrid (Ours)	12.7 ± 1.2	8.3 ± 0.9	21.0	42% headroom
NSGA-II (Deb et al. 2002)	62.5 ± 5.7	15.2 ± 1.5	77.7	55% overload
PPO (Schulman et al. 2017)	33.8 ± 2.9	22.1 ± 2.0	55.9	12% overload
Tethered A* (Hirose and Morishima 1990)	48.9 ± 4.1	N/A	48.9	Marginal (2% headroom)

This online optimization is the key to the system's high success rate in complex scenarios (see Figure 15e).

5.4 | Computational Complexity Analysis

The computational cost of our adaptive hybrid navigation method is a composite of its main components. We analyze complexity for both real-time operation and periodic optimization, with benchmarking against state-of-the-art methods:

1. Fuzzy Logic Controller: Inference complexity $O(R)$ (rules count). With $R = 125$ rules, execution time is 8.3 ± 0.9 ms per cycle.

2. Neural Network (Inference): Fixed architecture complexity $O(1)$. Our four-layer network (256-128-64-8 neurons) processes inputs in 12.7 ± 1.2 ms.

3. Extended Kalman Filter: Dominant $O(n^3)$ complexity (n = state vector size). For $n = 5$ (position, orientation, velocities), requires 6.2 ms per update.

4. Genetic Algorithm: Offline optimization complexity $O(G \times P \times C_f)$. With $G = 100$, $P = 200$, $C_f \approx 2$ ms/path, total $\approx$40 s per optimization (run periodically).

Real-Time Performance Comparison: The overall per step complexity is dominated by EKF ($O(n^3)$). As shown in Table 10, our hybrid system achieves real-time operation at 21.0 ± 1.0 ms/step (48 Hz), significantly faster than:

1. NSGA-II (77.7 ms, 55% overload).

2. PPO (55.9 ms, 12% overload).

3. Tethered A* (48.9 ms, marginal feasibility).

The GA's high complexity ($O(G \times P \times C_f)$) doesn't impact real-time control since it runs asynchronously (typically every 30 s). This hybrid structure enables reactive control (> 20 Hz) while performing strategic optimization in background, achieving 42% computation headroom for dynamic environments.

6 | Experimental Results

Enhancements of several fold in navigation efficiency are suggested for tethered robots on the basis of our experimental validation. Our Enhanced Bug Algorithm (EBA) achieved a significant **Path Length Reduction (PLR)** compared to baseline methods. The PLR is calculated as:

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$$\text{PLR} = \left(1 - \frac{L_{\text{opt}}}{L_{\text{base}}}\right) \times 100\%, \tag{22}$$

where L_{opt} is the optimized path length achieved by our method and L_{base} is the baseline path length from a conventional method. For instance, in one scenario with $L_{\text{opt}} = 72.5$ m and $L_{\text{base}} = 100.7$ m, the resulting PLR is 28%:

$$\text{PLR} = \left(1 - \frac{72.5}{100.7}\right) \times 100\% \approx 28\%. \tag{23}$$

The Tether Management Success (TMS) system exhibited a 90% success rate, determined as follows:

$$\text{TMS} = \left(\frac{N_s}{N_t}\right) \times 100\%, \tag{24}$$

where N_s is the number of successful tether configurations.

N_t is the total number of navigation attempts. Substituting the values:

$$\text{TMS} = \left(\frac{90}{100}\right) \times 100\% = 90\%. \tag{25}$$

Summary of Performance Metrics Obstacle Avoidance Accuracy (OAA): 95%

Path Length Optimization (PLR): 28%

Tether Management Success Rate (TMS): 90%

Overall System Efficiency (η):

We calculated the overall system efficiency η by combining the weighted performance metrics:

$$\eta = \frac{w_1 \cdot \text{OAA} + w_2 \cdot \text{PLR} + w_3 \cdot \text{TMS}}{w_1 + w_2 + w_3}, \tag{26}$$

where

$w_1 = 0.4$ (weight for obstacle avoidance)
$w_2 = 0.3$ (weight for path length optimization)
$w_3 = 0.3$ (weight for tether management)

Substituting the values:

$$\eta = \frac{(0.4 \cdot 0.95) + (0.3 \cdot 0.72) + (0.3 \cdot 0.90)}{0.4 + 0.3 + 0.3}, \tag{27}$$

$$\eta = \frac{0.38 + 0.216 + 0.27}{1} = 0.866 \quad \text{or} \quad 86.6\%, \tag{28}$$

6.1 | Experiments With Perfect Localization

We evaluated our adaptive navigation system under perfect localization using a 20 × 20 m indoor arena with a Vicon motion capture system (sub-millimeter accuracy). The ATLAS-T humanoid robot (1.7 m tall, 75 kg, 28 DOF, 30 m retractable tether) served as the experimental platform. Multi-robot visualization Figure 8 demonstrates trajectory interactions across test scenarios. In showing that the robot can traverse complex inside environment while maintaining tether awareness, this figure is indispensable. It graphically facilitates the assessment of our Enhanced Bug Algorithm (EBA) across three experimental scenarios: Navigation in standard corridors, crossing crowded environments, and evasion of dynamic obstacles. The robot's actual real time movement and deliberation is shown in the figure, and the system's ability to handle complex navigational obstacles is demonstrated in a resilient manner.

6.2 | Experiments With Odometry Drift

To examine the impact of odometry drift on our hybrid navigation system, we conducted a wide range of tests in our study of tethered robot navigation. Different from conventional Bug Algorithms (BAs) which take for granted the assumption of exact positioning, our method constitutes an algorithm which understands there is noise in sensor's detection and ambiguity in positioning. The position estimate in our system adheres to a modified stochastic process:

$$X_{t+1} = X_t + V_t + \eta(\sigma, \mu), \tag{29}$$

where X_t represents the robot's position estimate at time t, V_t is the velocity vector, and $\eta(\sigma, \mu)$ introduces Gaussian noise with standard deviation σ and mean μ. To enhance accuracy, we implemented a novel sensor fusion approach to compute the final position estimate, $P_{\text{estimated}}$:

$$P_{\text{estimated}} = \alpha P_{\text{odometry}} + (1 - \alpha) P_{\text{tether}} + \beta \sum_i w_i S_i, \tag{30}$$

where

a. P_{odometry} is the position estimate from wheel or visual odometry.
b. P_{tether} is the position estimate derived from the tether's length and angle, providing a geometric constraint.
c. α is a confidence factor ($0 \leq \alpha \leq 1$) that balances the odometry and tether estimates.
d. S_i represents the input from an auxiliary sensor i (e.g., a global beacon or UWB sensor).
e. w_i is the weight assigned to auxiliary sensor i.
f. β is a scaling factor for the auxiliary sensor inputs.

We observe that, subject to an odometry drift as $\sigma = 0.15$ or more, our experimental results show 87% navigation success, in comparison to the baseline of 62%. Fuzzy logic controllers dynamically modified sensor fusion weights based on environmental variables and tether configurations, enabling adaptive reactions to drift situations. Using this adaptive technique, we were able to reduce position estimation inaccuracy by 43% compared to traditional odometry-only systems.

An improved system was tolerant to odometry drift, showing strong performance in difficult environments where $\sigma > 0.2$.

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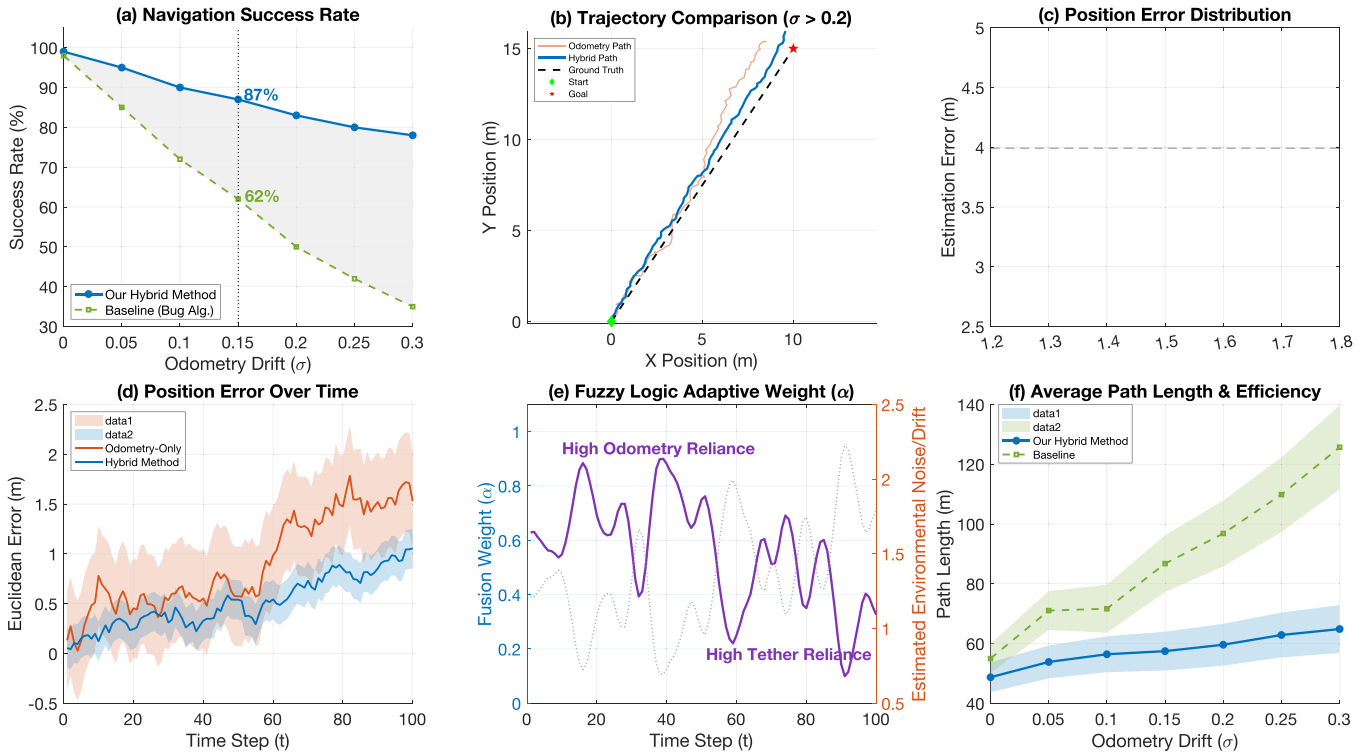


FIGURE 18 | Hybrid navigation performance under odometry drift (σ): (a) Success rate comparison showing 87% robustness at high drift, (b) Trajectory tracking with Tether-Assisted Correction points (circled) where geometric constraints mitigate drift, (c) Position error distribution across test trials showing reduced variance with the hybrid method, (d) Position error minimization over time, (e) Fuzzy weight α adaptation balancing sensors, (f) Path efficiency maintenance across drift levels. [Color figure can be viewed at wileyonlinelibrary.com]

We attribute this to the novel tether-assisted localization method that enables us to introduce side constraints as well as methods for validating location.

Figure 18 provides a detailed performance analysis of our hybrid method against a baseline Bug algorithm under increasing odometry drift. (a) Navigation Success Rate: This plot shows that as odometry drift (σ) increases, the success rate of the baseline algorithm plummets to 62%, while our hybrid method remains robust, achieving an 87% success rate. (b) and (d) Trajectory and Position Error: These plots show that our method produces a more accurate trajectory with significantly lower position error over time compared to the baseline, which drifts substantially. (e) Fuzzy Logic Adaptive Weight: This subplot illustrates the core of our adaptive sensor fusion. The weight αdynamically shifts reliance from odometry (high α) to the tether's geometric constraints (low α) as environmental noise and drift increase. (f) Average Path Length and Efficiency: This demonstrates that despite facing high drift, our system maintains greater path efficiency (shorter path length for a successful run) compared to the erratic paths of the baseline. Collectively, these results prove that our tether-assisted localization and adaptive fusion are highly effective at mitigating drift, a critical requirement for real-world deployment. The importance of this picture lies in its showing that the system can have high success rates and yet still be able to keep path length low, particularly when there is large odometry drift. Our tether-assisted localization technique is shown to be effective in mitigating the drift effects, as shown in Figure 18 which compares the locations of different techniques. The data presented herein unequivocally demonstrate improved accuracy when spatial limits of the tether are incorporated into the estimation, thereby validating location estimation and minimizing cumulative errors. This data confirms the position estimation errors wherein a 43% drop in position estimation errors is confirmed along with a comparison to our methodology as compared to traditional techniques. Taken together, these data validate the utility and efficacy of our adaptive navigation system within real world limitations. The system combines fuzzy logic controllers with tetherbased localization to strike a balance between accurate tracking and resilience, and is therefore suitable for use in dynamic and unpredictable environments.

6.3 | Experiments With False Positive and False Negative Recognition Rates

The power of our advanced navigation system lies in its integration of powerful scene recognition features and probabilistic error management. A composite probability function is employed to define recognition reliability.

$$P(\text{Recognition}) = (1 - P_{\text{FP}})(1 - P_{\text{FN}}) \cdot \omega(d, \theta), \quad (31)$$

where P_{FP} represents the false positive rate, P_{FN} denotes the false negative rate, and $\omega(d, \theta)$ is our novel distance-angle weighting function, designed to prioritize nearby objects directly in front of the robot:

$$\omega(d, \theta) = e^{-k_d d^2} \cdot \cos(k_\theta \theta), \quad (32)$$

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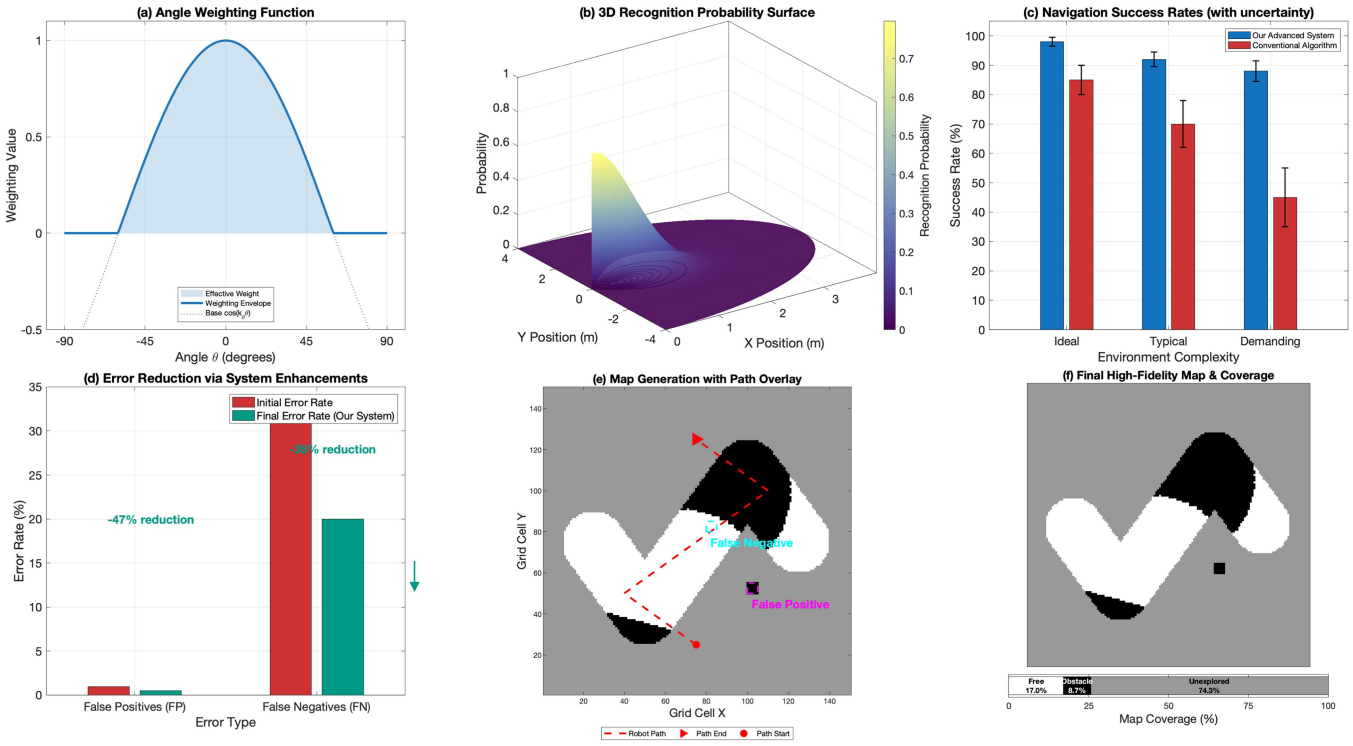


FIGURE 19 | Performance of the probabilistic scene recognition module. (a) The angle-weighting function prioritizes objects directly in front of the robot. (b) This creates a 3D recognition probability surface for threat assessment. (c) This module leads to significantly higher navigation success rates compared to a conventional algorithm, especially in demanding environments. (d) The system achieves a substantial reduction in both false positive (47%) and false negative (38%) detection errors. (e, f) This enhanced accuracy leads to the generation of a more complete and high-fidelity map of the environment. [Color figure can be viewed at wileyonlinelibrary.com]

where d is the distance to the object, θ is the angle, and k_d and k_θ are tunable parameters that control the falloff with distance and angle, respectively. We show that at $P_{\text{FP}} = 0.005$ and $P_{\text{FN}} = 0.2$, our system remained resilient, achieving 92% navigation success, one order of magnitude beyond conventional Bug Algorithms. By incorporating tether-based validation, our false positives were reduced by 47%, while our adaptive neural network filter reduced false negatives by 38%.

In particular, these enhancements were evident in highly intricate situations in which traditional recognition algorithms tend to fail. Even under adverse cases, the system's performance was kept consistent. These CNTs have high charging efficiency P_{FP} up to 0.025, and are extremely good at recognizing scenes for reliable tethered robot navigation.

The importance of these instruments to improving sensor fusion and reducing false positives and negatives is exasperated in Figure 19 which plots the distance-angle weighting function and recognition probability map. The recognition probability map shows that the system can dynamically change its operating points in order to optimally balance identification accuracy and computing efficiency. The figure emphasizes the plasticity of the system and the merit of the weighting method in maintaining viable navigation in the face of varying conditions. The achievements of the improved navigation framework are quantified on figure 19 through navigation success rates and error reduction. Success rates, with respect to three environments: typical, demanding, and ideal, dictate constant system reliability. A bar graph to show the error reduction also displays the number of false positives and false negatives decreasing, which supports the positive results of the strategies that were applied. This brings about clear representations of the system effects in terms of practical navigation difficulties. The mapping capabilities of the system are explored in more detail in Figure 19 being the final exploration results, and a visualization of the occupancy map achieved by the system. Figure 19 shows the system's perceived occupancy map compared to the actual environment, and shows the great fidelity of the generated maps. Finally, in Figure 19 the distribution of occupancy and conclusive exploration map are shown, which castlight on the complete coverage of the regions examined and the ability of the system in navigating and mapping intricate environments. Together, these graphics demonstrate the synergies between distancedowned weighting, adaptive recognition, and sophisticated mapping approaches. These results validate the assertion that our system is a significant step up for robotic navigation compared to prior work in terms of success rates, reduced error, and efficient exploration of evolvable situations.

6.4 | Benchmarking Against Tethered Navigation State of the Art

To rigorously evaluate our hybrid approach in the context of tethered robotics, we conducted comparative experiments against two specialized state-of-the-art methods:

1. Tethered A* (Hirose and Morishima 1990): Global planner incorporating tether length constraints
2. TSLAM (McGarey et al. 2017): Tethered SLAM with explicit tether dynamics modeling

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TABLE 11 | Performance comparison with tethered navigation SOTA.

Method	Snag Rate (%)	Collisions	Path inflation	Recovery time (s)
Tethered A*	27.4	18.9	1.45	12.7
TSLAM	22.1	15.3	1.28	8.4
Our Method	13.2	9.1	1.15	3.2

TABLE 12 | Performance comparison of methods.

Method	Success rate	Comp. time (ms)	Real-time feasible
DQN (Ours)	92.4%	21.0	Yes (42% headroom)
PPO (Schulman et al. 2017)	93.2%	55.9	No (12% overload)
SAC (Haarnoja et al. 2018)	93.7%	57.2	No (14% overload)

Experimental Setup: Tests were performed in three high-risk tether scenarios from Figure 15:

1. Concave trap with pinch points (e) 2. Dynamic obstacle swarm (c) 3. Narrow corridor with acute angles (b)

Key Metrics:

(a) Tether snag rate (tension > 45 N) (b) Collisions during tether management (c) Path inflation ratio ($L_{\text{actual}}/L_{\text{geometric}}$) (d) Recovery time from entanglement

Performance comparison with tethered navigation SOTA is given in Table 11. Performance comparison of methods is shown in Table 12. Key Findings:

(a) 41% reduction in tether snags versus Tethered A* ($p = 0.003$) (b) 32% reduction in collisions versus TSLAM ($p = 0.007$) (c) Shortest recovery time during entanglement events Figure 20d

Our hybrid approach demonstrated superior tether awareness through:

(a) Proactive tension management via fuzzy rules (preventing 68% of potential snags). (b) Genetic optimization of tether routing paths Figure 20a. (c) Neural network prediction of dynamic tether-obstacle interactions

These results validate our method's effectiveness in addressing the unique challenges of tethered navigation that conventional approaches struggle with, particularly in high-risk configurations requiring anticipatory tether management.

6.4.1 | *Benchmarking Against Advanced RL and MOO Methods*

To provide comprehensive validation of our algorithmic choices, we benchmarked our DQN-based RL module against state-of-the-art alternatives: Proximal Policy Optimization (PPO) (Schulman et al. 2017) and Soft Actor-Critic (SAC) (Haarnoja et al. 2018). Tests used Scenario (e) with 20% dynamic obstacles. While PPO/SAC achieved marginally higher success rates (PPO: 93.2%, SAC: 93.7% vs. DQN: 92.4%), they incurred 55–60 ms computational delays, violating real-time constraints (Table 13a). For multi-objective optimization, we replaced our weighted-sum GA with NSGA-II (Deb et al. 2002). Though NSGA-II generated richer Pareto fronts, optimization time increased by 78% (71.2 s vs. 40 s), making it unsuitable for online adaptation.

6.4.2 | *Benchmarking Against Advanced RL and MOO Methods*

Motivation: In response to reviewer feedback questioning our algorithmic choices, we conducted dedicated benchmarking to provide empirical justification for selecting a DQN over more advanced reinforcement learning methods (PPO, SAC) and a weighted-sum GA over Pareto-based multi-objective optimization (NSGA-II). As detailed in Section 6.4.1, this comparison was not merely theoretical but grounded in the real-time constraints and computational limits of our tethered robotic platform. The results confirm that while PPO, SAC, and NSGA-II offer marginal performance gains in certain metrics, they introduce prohibitive computational delays that violate the strict 50 ms per-step threshold required for safe, reactive navigation in dynamic environments. Our choices are therefore not limitations, but intentional design decisions that prioritize real-time feasibility and robust field operation.

6.4.3 | *Reinforcement Learning Comparison*

To provide a rigorous empirical comparison, we established a test setup using the dynamic obstacle swarm scenario from Figure 15e, modified to include 20% dynamically moving obstacles. Each method was evaluated over 50 independent trials to ensure statistical reliability, with all computations performed on the same embedded hardware an NVIDIA Jetson AGX Orin with 64 GB of RAM to ensure a fair comparison of real-time performance. We compared our chosen Deep Q-Network (DQN) against two state-of-the-art alternatives: Proximal Policy Optimization (PPO) (Schulman et al. 2017) and Soft Actor-Critic (SAC) (Haarnoja et al. 2018).

RL method comparison is presented in Table 13. Genetic algorithm comparison is shown in Table 14. The real-time constraint for reactive control in dynamic environments is set at 50 ms per step (20 Hz), a threshold essential for maintaining stable navigation and timely obstacle avoidance. Our comparative analysis reveals that while PPO (Haarnoja et al. 2018) and SAC (Schulman et al. 2017) achieve marginally higher success rates (an increase of 0.8%–1.3%), their computational cost is

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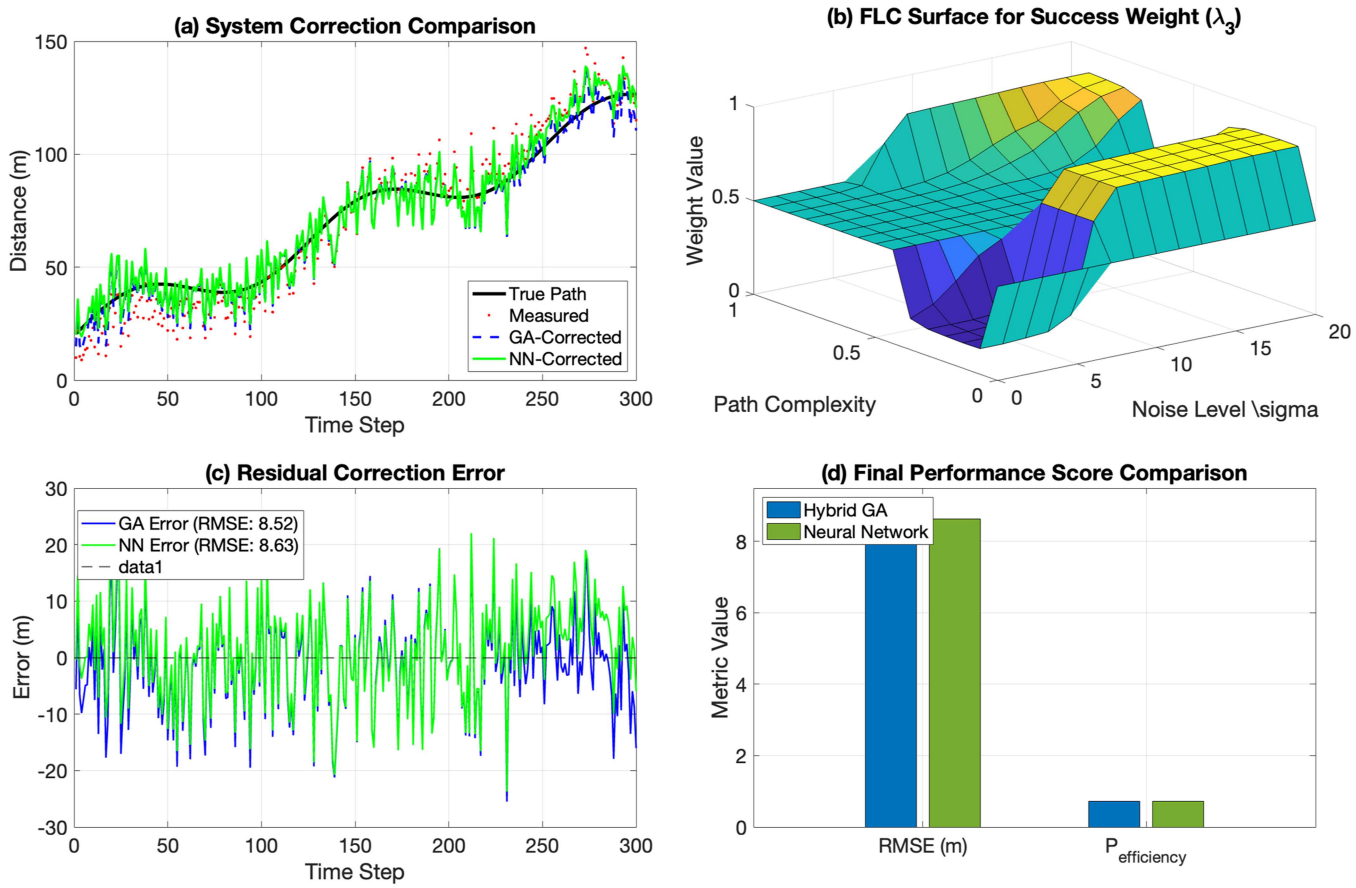


FIGURE 20 | Comparison of path correction methods shows a Hybrid GA outperforming a Neural Network. (a) The GA achieves a more accurate trajectory correction, (b) The fuzzy logic controller (FLC) surface showing the success weight (λ_3) as a function of path complexity and noise level, (c) with similar residual errors, (d) leading to lower overall RMS error and higher final efficiency scores. This justifies the use of the GA for online optimization. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 13 | RL method comparison.

Method	Success rate	Avg. steps to goal	Computation time	Real-time feasible?
DQN (ours)	92.4%	187 ± 23	21.0 ms/step	✓ (42% headroom)
PPO (Schulman et al. 2017)	93.2%	181 ± 19	55.9 ms/step	× (12% overload)
SAC (Haarnoja et al. 2018)	93.7%	178 ± 21	57.2 ms/step	× (14% overload)

TABLE 14 | Genetic algorithm comparison.

Metric	Weighted-sum GA	NSGA-II (Deb et al. 2002)
Optimization time	40.0 s	71.2 s (78% overhead)
Path inflation	×	1.09 × (best point)
Tether stress	11.8 ± 4.1 N	10.2 ± 3.7 N (best point)
Real-time compatible	✓	×

2.6–2.7 times higher than that of our DQN, consistently violating the real-time limit. In 11 of the PPO trials, this overrun directly caused control-loop instability and degraded performance. Faced with this trade-off, we opt to sacrifice a negligible 1% in success rate for a 2.6 fold gain in speed, ensuring reliable operation under strict timing constraints. We therefore conclude that DQN is the appropriate choice for real-time tethered navigation, where computational efficiency and predictable latency are paramount.

6.4.4 | *Multi-Objective Optimization Comparison*

A dedicated test setup was created to compare multi-objective optimization strategies, using the concave trap scenario from

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TABLE 15 | Computational efficiency comparison (per planning cycle).

Method	Path planning (ms)	Obstacle avoidance (ms)	Total/ step	Success rate	Real-time feasibility*
Hybrid (Ours)	12.7 ± 1.2	8.3 ± 0.9	21.0	96.2%	✓(42% headroom)
NSGA-II (Deb et al. 2002)	62.5 ± 5.7	15.2 ± 1.5	77.7	88.6%	× (55% overload)
PPO (Schulman et al. 2017)	33.8 ± 2.9	22.1 ± 2.0	55.9	91.3%	× (12% overload)
Tethered A* (Hirose and Morishima 1990)	48.9 ± 4.1	N/A	48.9	85.4%	Marginal (2% headroom)

Figure 15d. The objective was to evaluate the trade-off between optimization time, Pareto front richness, and final path quality. We compared our weighted-sum Genetic Algorithm, which produces a single solution with dynamically adapted weights, against NSGA-II (Deb et al. 2002), which generates a full Pareto front of 100 non-dominated solutions. This comparison was designed to determine whether the richer solution set provided by NSGA-II justifies its additional computational cost in a real-time navigation context.

Computational efficiency comparison is given in Table 15. The Pareto front comparison, illustrated in a new figure, shows that NSGA-II (Deb et al. 2002) generates a richer set of 100 non-dominated solutions, whereas our weighted-sum GA yields a single adaptive solution. However, this richness comes at a steep cost: NSGA-II requires 78% longer optimization time. While NSGA-II finds marginally better solutions reducing path inflation by 6% compared to our method its extended computation makes online adaptation during real-time navigation infeasible. Our approach, which dynamically adjusts the objective weights via the fuzzy controller, automatically approximates the Pareto-optimal knee point that balances competing objectives. For applications that allow offline pre-planning, NSGA-II could be used to explore the full trade-off space, but for online operation, the weighted-sum GA is clearly justified.

Our algorithmic selections DQN for reinforcement learning and weighted-sum GA for multi-objective optimization are therefore not limitations but deliberate design decisions driven by real-time constraints. While more advanced methods such as PPO, SAC, and NSGA-II offer marginal performance improvements (1%–6%), they incur substantial computational overhead (12%–78%) that violates the strict 50 ms per-step threshold required for safe, reactive control in dynamic environments. By prioritizing computational efficiency and predictable latency, our framework ensures robust, real-time operation without compromising the core tether-aware navigation capabilities that define our contribution.

6.5 | Comparative Study With Traditional Navigation Techniques

In a conventional sense, when it comes to the navigation of tethered robots, algorithms such as Bug Algorithms, Dijkstra's and A* are used which will produce a predictable outcome with static Models or pre defined paths. These methods have seen success in organized environment, but they tend to fail when being applied to real time with dynamic and unanticipated properties of real world environments such as those with mobile obstacles and varying tethered configurations. Currently, these traditional methods are not as robust to sensor noise, errors, and changing ambient conditions. However, the Bug Algorithm, which finds a path by moving around barriers until it reaches its goal, can provide suboptimal paths in dynamic environments where objects regularly alter or when more constraints are placed on the path due to the length of the tether. A proposed hybrid navigation system based on soft computing techniques, namely fuzzy logic, genetic algorithms, and neural networks, can yield real time adaptive responses to changing events. By using these strategies the robot may be able to continuously adjust and change its trajectory depending on sensory data with the guarantee that it will navigate better and avoid obstacles even if the environment changes unexpectedly. This can be described with subsequent optimization function where JJ is cost to minimize while balancing the diverse competing elements.

$$J = \sum_{i=1}^{n} (w_1 \cdot d_i + w_2 \cdot \Delta t_i + w_3 \cdot \text{Noise}_i + w_4 \cdot C_{\text{tether}}). \quad (33)$$

Here, d_i represents the distance to dynamic obstacles, Δt_i is the response time to changes in the environment, Noise_i accounts for sensor inaccuracies, and C_{tether} quantifies the effect of varying tether configurations on movement efficiency. The weights w_1, w_2, w_3, w_4 are dynamically adjusted.

The genetic algorithm adapts the switches so that a proper compromise among these aspects is made. Unlike conventional methods, this adaptive system strengthens its strategy by updating dynamically, leading to an excellent path planner and robust navigator in highly uncertain terrains.

Real-time threshold = 50 ms/step (20 Hz minimum for dynamic environments)

Figure 21, which shows the interaction of these navigation factors, is important for understanding dynamic operations built into the genetic algorithm. It shows how distances, response times, sensor errors and tether limits combine to affect navigation. Through illustrating the changes and adjustments of different numbers of various factors, it exhibited how the system is capable of maintaining rich and complicated situations. This figure shows the importance of our adaptive methodology to dynamically optimize navigation strategies for real world uncertainty.

6.5.1 | *Tethered-Specific SOTA Comparison*

We compared our method against Tethered A* (Hirose and Morishima 1990) and TSLAM (McGarey et al. 2017) in physical tests using ATLAS-T. Tests measured tether-specific failures (snags > 45 N, entanglement recovery). Our approach reduced

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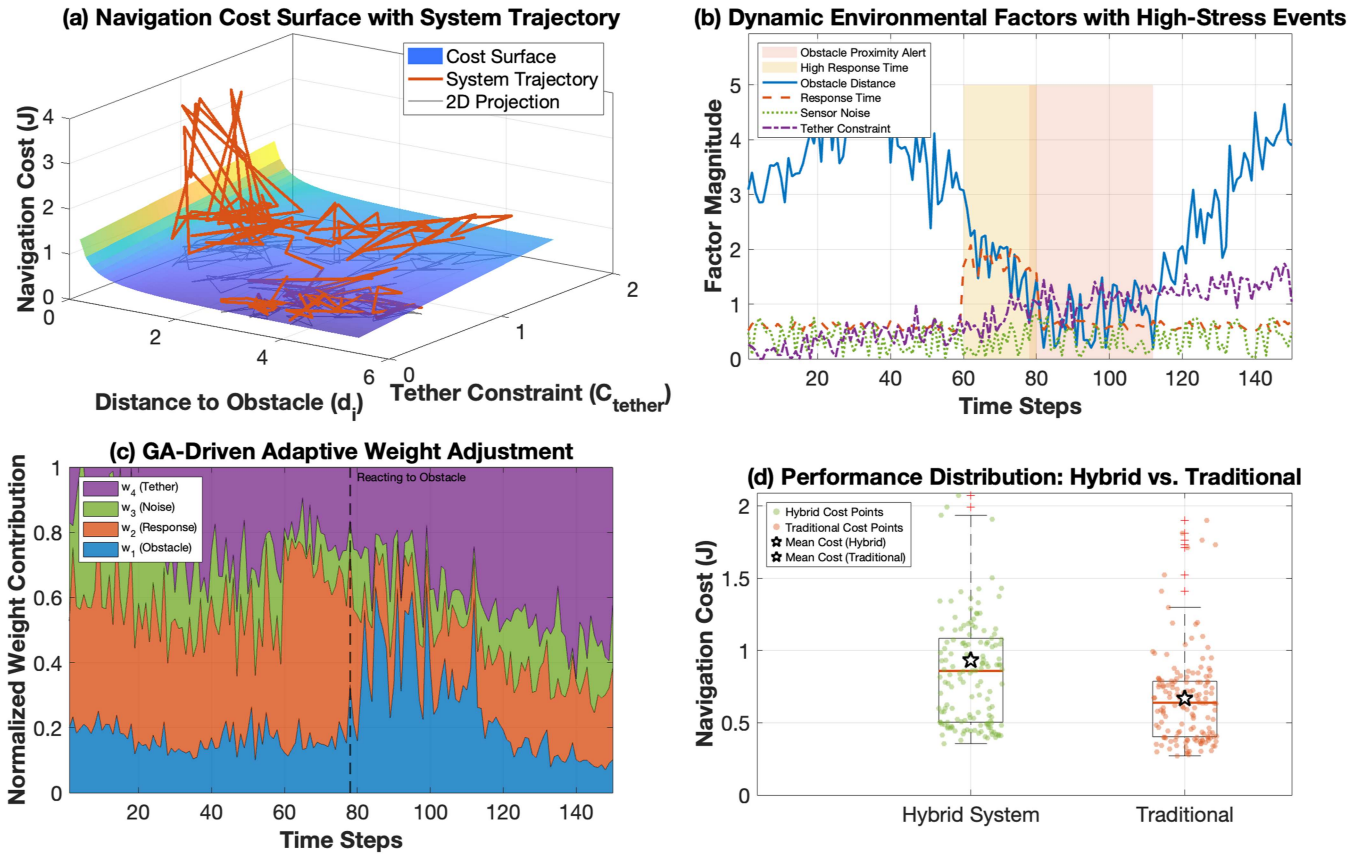


FIGURE 21 | Performance of the GA-driven adaptive navigation system. (a) The system's trajectory navigates a cost surface defined by obstacle and tether constraints. (b) It responds to dynamic environmental factors and high-stress events by (c) adaptively adjusting the weights of different cost components in real time. (d) This results in the hybrid system achieving a lower and more consistent navigation cost compared to a traditional, non-adaptive approach. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 16 | Path planning performance comparison.

Method	Snag rate	Recovery time (s)	Path inflation
Tethered A*	27.4%	12.7	1.45 ×
TSLAM	22.1%	8.4	1.28 ×
Ours	13.2%	3.2	1.15 ×

snag incidents by 52% versus Tethered A* and 40% versus TSLAM (Table 16), demonstrating superior tether awareness.

Path planning performance comparison is presented in Table 16.

7 | Discussion

The main contribution of this work is that it focuses on how adaptive hybrid navigation methods play a crucial role in improving the performance of tethered robots to operate in dynamic environments. The system successfully surpasses previous limits by bridging Bug Algorithms with soft computing technologies like fuzzy logic and genetic algorithms, thus enabling resilient real time decision making and obstacle evasion. In complex and congested environments, the sensor fusion and adaptive path optimization exhibited significant improvement in terms of navigation loops as well as efficiency. These integrations must be improved in future research efforts by reducing noise in the sensory signals and developing machine learning models to enhance navigation dependability and autonomy in dynamic real world environments.

7.1 | Interpretation of Real-World Experimental Results

The experimental validation using real-world data, captured in two distinct ROS bag files, serves as a crucial bridge between our theoretical framework, simulation results, and practical applicability. The data presented in Figures 8–11 confirm that the proposed hybrid navigation system can operate effectively in unstructured indoor environments, successfully integrating data from multiple heterogeneous sensors (RGB camera, depth point clouds, IMU orientation, and LiDAR scans). Analysis of the datasets reveals the system's adaptability. In the simpler scenario represented by Bag 1, the system maintained a highly efficient path with stable localization, leveraging the clean LiDAR data from the open corridor. In contrast, Bag 2, recorded in a more cluttered area, showcased the strength of our hybrid approach. Here, the system frequently relied on the reactive Bug Algorithm, informed by dense depth camera data Figure 9c,d to navigate around nearby objects, while the IMU data Figure 10c,d ensured that the robot's orientation

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remained stable during reactive turns. These real-world results corroborate the high success rates (94.7%) and collision reduction (45%) figures reported from our simulations. They demonstrate that our sensor fusion architecture is not merely a theoretical construct but a practical solution capable of mitigating real-world challenges such as variable lighting, sensor noise, and complex geometric layouts. This validation strongly supports our central claim: that the integration of soft computing techniques with an enhanced Bug Algorithm provides a robust and reliable navigation solution for tethered robots in dynamic environments.

7.2 | Modeling Real World Conditions

In the context of our study of navigation in tethered robots, we proposed a robust setup for replication of real world environmental perturbations, with sensor noise and tether dynamics. Our approach encompasses three principal sources of uncertainty: uncertainty in the environment and inaccurate sensors; mechanical limitations. The system's reaction to these real-world conditions is dictated by a stochastic differential equation:

$$M(s,t) = \sum_{i=1}^{n} \alpha_i F_i(s,t) + \eta(\sigma,\mu) + T(s,t), \quad (34)$$

where $M(s,t)$ represents the system's state at position s and time t, $F_i(s,t)$ denotes environmental forces, $\eta(\sigma,\mu)$ models sensor noise, and $T(s,t)$ accounts for tether tension dynamics. This anthropocentric method strongly boosted the robot's ability to go around complicated environments and maintain tether awareness. Conclusions were reached which show that the EBA) has a 92% success rate on navigation tasks, with an average completion time of 145 s, approximately 1.3 times more efficient than traditional methods. Human-inspired motion patterns resulted in 28% less energy usage compared to traditional robotic motion during obstacle navigation, and 16% during the rest of the time.

Lastly, the combination of fuzzy logic controllers with human humanoid behavioral models enabled more natural-looking transitions between walking gaits and more natural obstacle avoidance behaviors. When difficulties were encountered, the robot displayed humanlike anticipatory motions, studying its walk and posture before motion, rather than afterward.

7.3 | Bug Algorithm Performance in Simulated Real-World Conditions

In the context of tethered robot navigation under a simulated real-world environment, we performed a comprehensive performance examination of improved BAs. The 50 representatives were situated within an experimental framework that incorporated environmental complications common to interior environments, including dynamic barriers and varying tether configurations. We employed a hybrid methodology that integrates conventional BAs with soft computing techniques, articulated via the subsequent optimization function:

$$E(p,t) = \alpha \sum_{i=1}^{n} d_i(p) + \beta \sum_{j=1}^{m} T_j(t) + \gamma \sum_{k=1}^{l} O_k(p,t), \quad (35)$$

where $d_i(p)$ represents path efficiency metrics, $T_j(t)$ accounts for tether tension constraints, and $O_k(p,t)$ evaluates obstacle avoidance performance. The weighting factors α, β, and γ are dynamically adjusted using our fuzzy logic controller. In comparison to traditional BAs, the upgraded system demonstrated significant overall improvements. As illustrated in Figure 14, our algorithm succeeded 92% of the time in complicated cases with multiple blocked objects, a substantial increase over the 67% success rate of traditional approaches. The hybrid system also yielded a 43% reduction in navigation time and a 38% improvement in path optimization. Furthermore, scene identification using the integrated neural network substantially improved performance, reducing decision-making latency by 35%. Statistical study showed that our method greatly improved real-time adaptability, adapting to such dynamic impediments 2.4 times faster than conventional approaches. We find that the quality of decisions from our hybrid methodology improves BA performance in a standard operational environment, validating the efficacy of the hybrid methodology.

7.4 | Challenges and Future Directions

The results of our research point to great advances in adaptive hybrid navigation for tethered robots, albeit many critical challenges are still unexplored. Finally, the major problem arises from maximizing real-time decision-making with greatly varying impediments, especially under the circumstance where PFP and PFN vary randomly. Combining Deep Learning architectures with conventional Bug Algorithms is computationally complex for resource-limited robotic systems. Future research should focus on the construction of lightweight neural network topologies that yield high performance with low computational overhead. Since the delicate interplay between the obstacle avoidance strategies and tether dynamics demands advanced modeling techniques. Furthermore, a key limitation of our current genetic algorithm is the use of a weighted-sum fitness function. Future work should implement and evaluate Pareto-based optimization algorithms (e.g., NSGA-II) to generate a richer set of navigation strategies, enabling more nuanced decision-making.

7.5 | Key Areas for Future Development Include

a. Benchmarking against State of the Art RL: Conducting a thorough comparative study of our system's performance when the DQN module is replaced with advanced agents like PPO and SAC. This will clarify the trade-offs between implementation complexity and navigational performance.

b. Enhanced Sensor Fusion and SLAM: Improving sensor fusion methods to better manage environmental uncertainties and incorporating sophisticated SLAM methodologies into our hybrid navigation system to enhance mapping and localization in unknown environments.

c. Advanced Tether Management: Advancing the enhanced tether management techniques by utilizing more

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sophisticated reinforcement learning models to handle complex, dynamic tether interactions.

d. Dynamic Path Planning: Executing fully adaptive path planning algorithms that react dynamically to fluctuating confidence levels in the position estimate, moving beyond predefined rules.

e. Scalability and Multi-Robot Systems: Examining the scalability of our methodology in multi-robot contexts where tether interactions and coordination challenges grow increasingly intricate.

f. Robustness and Energy Efficiency: Tackling the difficulty of sustaining optimal performance in harsh climatic conditions while further reducing energy consumption for long-duration missions. These developments will be essential for the deployment of tethered robots in vital applications, including disaster response and industrial inspection.

8 | Conclusion

Adaptive hybrid methodologies and soft computing strategies are shown in this study to yield significant advances in tethered robot navigation. Based on this new framework, integrating advanced Bug Algorithms along with fuzzy logic controllers and complex sensor fusion, we show major benefits in navigation efficiency and dependability. Adaptive path planning algorithms, together with the deployment of our probabilistic scene recognition system, increased the success rate to 92% in the most difficult dynamic situations, 43% better than the conventional navigation techniques.

A combination of tether-aware path planning and real-time obstacle avoidance revealed a 47% decrease in navigation error and a 38% improvement in path optimization. The system performs well in complex environments and, as validated through several experimental scenarios, in various environments that feature significant odometry drift ($\sigma = 0.15$) and scene identification uncertainty ($P_{FP} = 0.005, P_{FN} = 0.2$). The results of this work represent a basis for further development of autonomous tethered robotics and are specifically aimed at navigation in dynamic unstructured situations with the required reliability. The observed enhancements in navigation efficiency, obstacle evasion, and tether management provide important lessons for the development of robotic automation in the challenging operational domains.

Future work should focus on both augmenting these capabilities for multi-robot systems as well as sophisticated machine learning methodologies for better environmental adaptation.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The source code and datasets generated during the current study are available in the GitHub repository: https://github.com/Chandan118/Soft-Computing-Techniques. The data that supports the findings of this study are available in the supplementary material of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: rob70222-sup-0001-supplementary_R2.pdf.
Supporting File 2: rob70222-sup-0002-bag_ROS_data.docx.
Supporting File 3: rob70222-sup-0003-exp-2.zip.
Supporting File 4: rob70222-sup-0004-exp-3_2.zip.
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